Unit-2
Applications of Diodes and Rectifiers

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2.1 BASIC OF DIODE :

PN junction diode is widely used in rectifier circuits. It is also used in clamper, clipper and logic circuits. In addition to the PN junction diode, other devices are also used in different circuits. These devices include zener diode, varactor diode, photo diode, light emitting diode, thermister, pin diode etc. In this chapter first we shall study different rectifier circuits and clipper and clipper circuits using PN junction diodes and then we shall study the construction, characteristics and applications of the terminal devices mentioned above.

2.1.1 N-Type Semiconductor :

When in pure germanium or silicon the impurity of the pentavalent element (having five valence electrons) like arsenic, antimony or phosphorous is added in very small definite proportion, N-type semiconductor results. Two dimensional picture of N-type semiconductor is shown in Fig. 2.1.

In the figure, one antimony atom has taken place of one silicon atom. Antimony atom has five valence electrons. Out of these, four electrons will be joined to four electrons by the covalent bonds. But one electron is left out which does not have any covalent bond. Moreover it is away from the nucleus. So it is attracted towards the nucleus with lesser force of attraction. So very small amount of energy is needed to make it free. For example only 0.02 eV is needed in case of germanium and 0.05 eV energy is required in case of silicon. This energy is sufficient to bring this electron to the conduction band. This energy is so small that it is available at the room temperature.

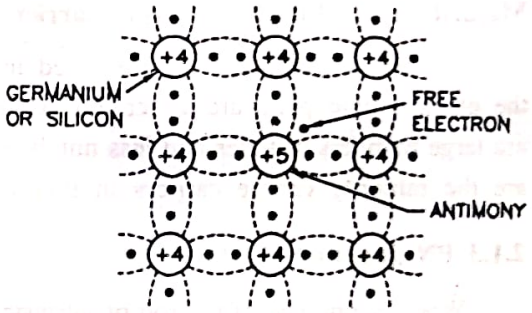


FIG. 2.1 : N-TYPE SEMICONDUCTOR

Thus the impurity element gives that number of free electrons equal to the number of the impurity atom. This impurity element is called the donor impurity element. And such donated electrons are called the excess electrons.

Majority and Minority Charge Carriers in N-type Semiconductor :

In N-type semiconductors, electrons equal to the number of impurity atoms are available in the conduction band. Moreover at room temperature some covalent bonds are .broken and some electron-hole pairs are generated. So the total number of electrons is more compared to the number of holes. **So in N-type semiconductor electrons are the majority charge carriers and holes are the minority charge carriers.**

2.1.2 P-Type Semiconductor :

When in pure germanium or silicon, impurity of the trivalent element (boron, gallium or indium) having three valence electrons is added in small definite proportion, P-type semiconductor results. This type of semiconductor is shown in Fig. 2.2.

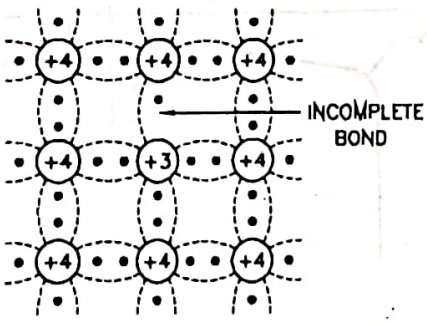


FIG. 2.2 : P-TYPE SEMICONDUCTOR GALLIUM ADDED TO SILICON

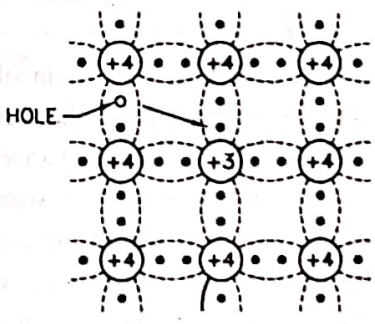


FIG. 2.3 : CREATION OF HOLE

In figure one gallium atom is shown surrounded by silicon atoms. Gallium has 3 valence electrons, so three silicon atoms can form the covalent bonds, but the fourth silicon atom can not. Thus there is deficiency of one electron in gallium atom. However this gap can be filled up by electron from the neighbouring atom. And this probability is more as it requires energy of only 0.01 eV which is available at the room temperature.

When gap in the atom of gallium is filled by the electron from the neighbouring silicon atom, a gap is produced in that silicon atom. Since it has lost an electron the net charge of that atom becomes positive and it is called 'hole'.

There are now four valence electrons instead of three in gallium atom. The covalent bond is complete. Its net charge becomes negative. This negative ion is bound with covalent bond in the crystal, so it is immobile.

Boron, indium and gallium are called the acceptor type impurities as this impurity element accepts electron from the crystal and becomes negative ion. Thus the holes are available as charge carriers in P-type semiconductor.

Majority and Minority Charge Carriers in P-type Semiconductor :

As seen above holes are produced in P-type semiconductor and act as charge carriers. In addition to this the electron-hole pairs are generated at room temperature because of breaking of covalent bonds. Hence there are large numbers of holes and less numbers of electrons. Hence holes are the majority charge carriers and electron are the minority charge carriers in P-type semiconductor.

2.1.3 PN Junction Diode :

When at one end of the rod of intrinsic semiconductor the acceptor type impurity is added P-type semiconductor is formed at that end. At the other end the acceptor type impurity is added. N-type semiconductor is formed. Thus in a semiconductor when there is P-type semiconductor at one end and N-type semiconductor near to it at the other end is called the PN junction diode.

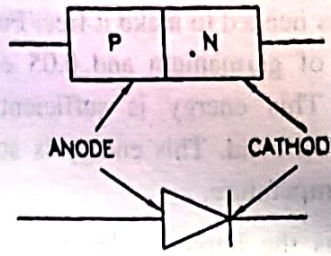


FIG. 2.4 : CONSTRUCTION AND SYMBOL OF PN JUNCTION DIODE

One terminal is brought out of the P-type and is called the anode. Similarly one terminal is brought o of the N-type and is called the cathode. Construction and symbol are shown in Fig. 2.4. The arrow shows th easy flow of current when it is forward biased.

2.1.4 PN Diode Characteristics :

The VI characteristic looks like that shown in Fig. 2.5.

Forward characteristic :

Looking to the forward characteristic it is seen that initially when the voltage across the diode is less than the knee voltage, the current is very very small because the barrier potential which is 0.7 V in silicon diode opposes the applied voltage. When the forward voltage reaches 0.7 V, the potential barrier is overcome and the junction offers very low resistance. Then if the forward voltage is increased the current rises rapidly. The increase in current is more compared to the increase in the voltage i.e. it does not obay the Ohm's law.

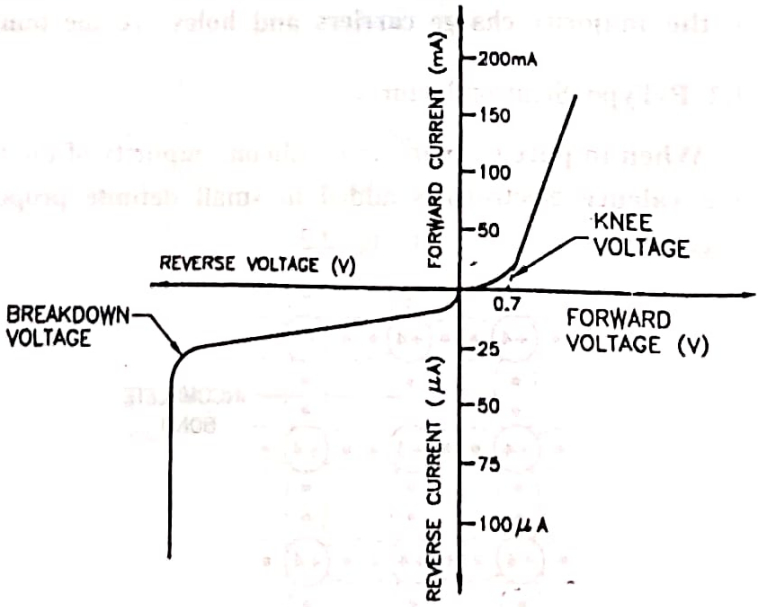


FIG. 2.5 : VI CHARACTERISTIC OF JUNCTION DIODE

Knee voltage :

The forward voltage at which the current through the forward biased diode starts to increase rapidly is known as the knee voltage. This value is 0.7 V for the silicon diode and 0.3 V for the germanium diode.

Reverse characteristic :

The reverse characteristic is also shown in Fig. 2.5. It is drawn in the fourth quadrant as the voltage is negative and current flow is also in the negative direction. Reverse voltage is in volts where as the current is in microampere. The reverse current is very small and independent of the applied reverse voltage, if the temperature is constant (the reverse current is temperature dependent). The current rises rapidly when the breakdown voltage is reached. The reverse current is limited by the current limiting resistor R.

Breakdown voltage :

The reverse voltage at which the junction breaks down and results in sudden increase in reverse current is known as the breakdown voltage.

2.1.5 Diode Applications :

There are many applications of PN junction diode. Some applications are listed here.

1. As low power and high power rectifier.
2. As a switch.
3. In clipping circuit.
4. In clamping circuit.
5. For protection of transistor, SCR and OP-Amp.
6. As detector in AM radio receiver (here point contact diode is used.)

2.1.6 Diode Clipping Circuits :

An important application of the PN junction diode is clipper and clamper. We shall now study the clipping and clamping circuits using junction diode.

Clipping circuits :

A clipping circuit is that with the help of which the undesired portion of a waveform is clipped off or removed without changing the shape of the other part of the waveform. The clipping circuit is also called the voltage or current limiter, slicer or the amplitude selector. Clipping circuit employs diode and resistor. Battery is used with the biased type clipper. When the flat waveform at the clipping level is desired, value of resistance R is chosen such that $R = \sqrt{R_f R_r}$, where R_f and R_r are the forward and reverse resistances of the diode. Following four types of clipper circuits are available :

1. Positive clipper.
2. Negative clipper.
3. Biased clipper and
4. Combined clipper

2.1.6.1 Positive Clipper :

The positive clipper is a circuit which removes or clips off the positive half cycles of the input a.c. waveform and gives out only the negative half cycles at the output. There are two types of circuits for the positive clipper.

These are :

1. Series positive clipper and
2. Shunt positive clipper.

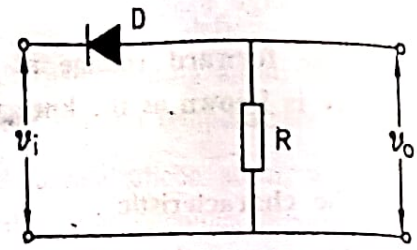


FIG. 2.6 : SERIES POSITIVE CLIPPER

1. Series positive clipper :

Circuit of series positive clipper is shown in Fig. 2.6. In this the diode is connected in series as shown. So this circuit is called the series positive clipper. The resistor *R* is connected in shunt and the output is taken across this resistor.

When the input voltage is positive, the diode is reverse biased and acts as switch open. So positive half cycles of a.c. input can not appear at the output. In other words all the input voltage during positive half cycle is dropped across the diode and output voltage is zero.

During the negative half cycle, the diode is forward biased and acts as a switch closed. So the whole negative half cycle is available at the output terminals. The waveforms are shown in Fig. 2.7. It is seen that the positive half cycles are clipped off.

2. Shunt positive clipper :

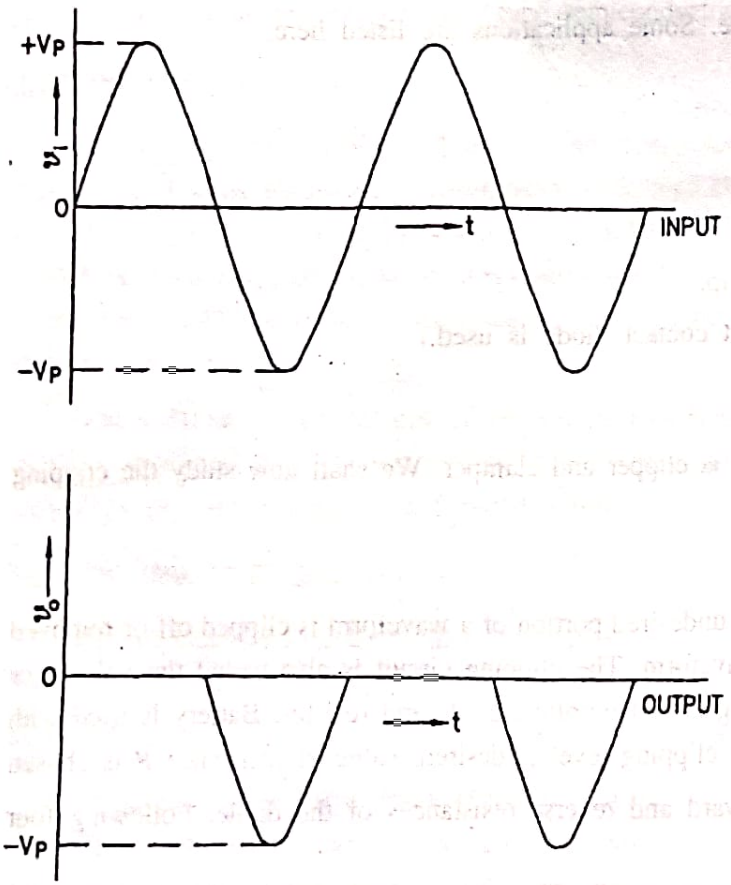


FIG. 2.7 : WAVEFORMS OF THE SERIES AND SHUNT TYPE POSITVE CLIPPER

In shunt positive clipper the diode is connected in shunt i.e. in parallel with the polarity as shown in Fig. 2.8.

During the positive half cycles of a.c. input, the diode is forward biased and acts as switch closed. So the whole input voltage is dropped across the resistor and the output voltage is zero.

During the negative half cycles, diode is reverse biased so acts as switch open. So the negative half cycles are available at the output. Thus positive half cycles are clipped. The input and output waveforms are shown in Fig. 2.7.

It is seen that the circuits behave as half wave rectifier.

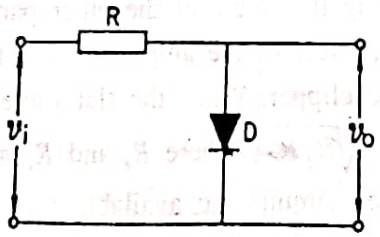


FIG. 2.8 : SHUNT POSITIVE CLIPPER

2.1.6.2 Negative Clipper :

In the negative clipper circuits, the negative half cycles of a.c. input are clipped off and only positive half cycles are available at the output. In this also there are two types of circuit configurations :

1. Series type and
2. Shunt type

1. Series negative clipper :

The circuit of series negative clipper is shown in Fig. 2.9. It is similar to the series positive clipper except that the polarity of the diode is reversed.

During the positive half cycles of a.c. input, the diode is forward biased and acts as switch closed. So the positive half cycles are available at the output. But during the negative half cycles the diode is reverse biased and acts as switch open. So zero voltage is available at the output. In other words, whole the negative input voltage is dropped across the reverse biased diode. The input and output waveforms are shown in Fig. 2.10.

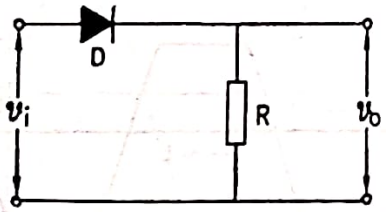


FIG. 2.9 : SERIES NEGATIVE CLIPPER

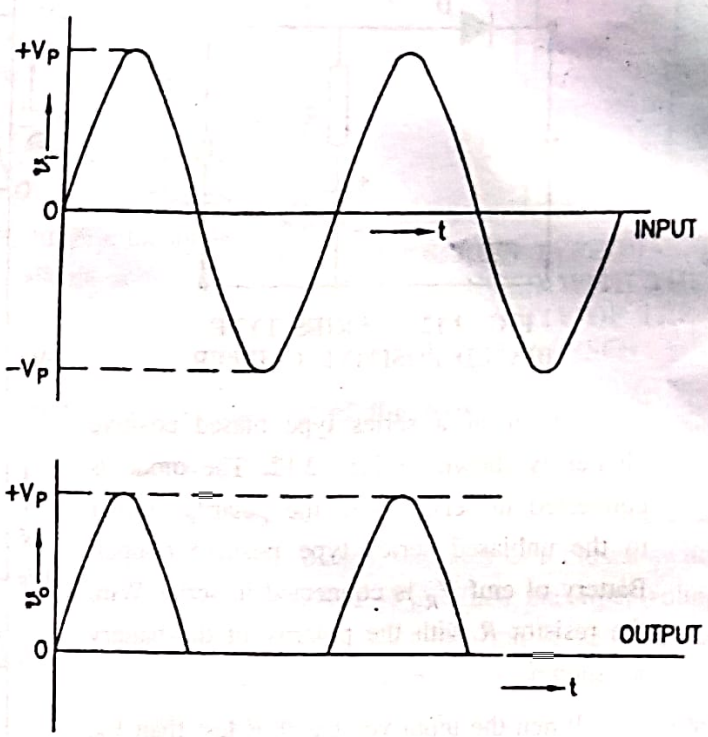


FIG. 2.10 : WAVEFORMS OF THE SERIES AND SHUNT TYPE NEGATIVE CLIPPER

2. Shunt negative clipper :

The circuit of shunt negative clipper is shown in Fig. 2.11. In this the diode is connected in shunt similar to the shunt positive clipper with the polarity of the diode reversed.

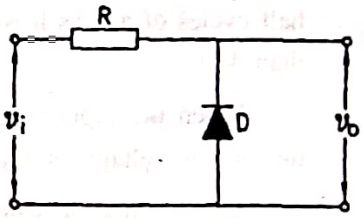


FIG. 2.11 : SHUNT NEGATIVE CLIPPER

During the positive half cycles of a.c. input, the diode is reverse biased so the diode acts as switch open, so the positive half cycles are available at the output. But during the negative half cycles the diode is forward biased and acts as short. So output becomes zero. All the input voltage is dropped across the resistor. Thus the negative half cycles are clipped off. The waveforms are shown in Fig. 2.10.

2.1.6.3 Biased Clipper :

Sometimes It becomes necessary to have circuits by which only some portion of the waveform has to be clipped off instead of the entire positive or negative half cycles. In such case the biased clipper circuits are employed. In biased clipper the desired level is obtained by using a battery in series with the resistor or diode depending upon the circuit configuration. Biased clippers are also of the two types :

(a) Biased positive clipper and

(b) Biased negative clipper

(a) Biased positive clipper :

Similar to the unbiased positive clipper the biased positive clippers are also of two types :

(1) Series type and

(2) Shunt type

(1) Series type biased positive clipper :

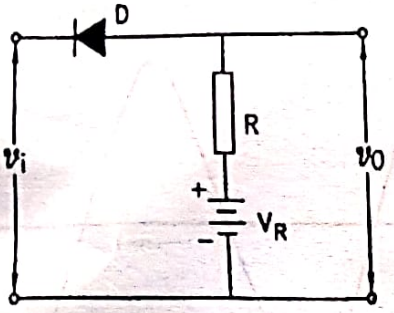


FIG. 2.12 : SERIES TYPE BIASED POSITIVE CLIPPER

Circuit of a series type biased positive clipper is shown in Fig. 2.12. The diode is connected in series With the polarity similar to the unbiased series type positive clipper. Battery of emf V_R is connected in series With the resistor R with the polarity of the battery as shown.

When the input voltage ϑ_i is less than V_R, the diode is forward biased and acts as switch closed. So input voltage appears at the output terminals (This includes the negative half cycles of a.c. as it is algebraically smaller than V_R).

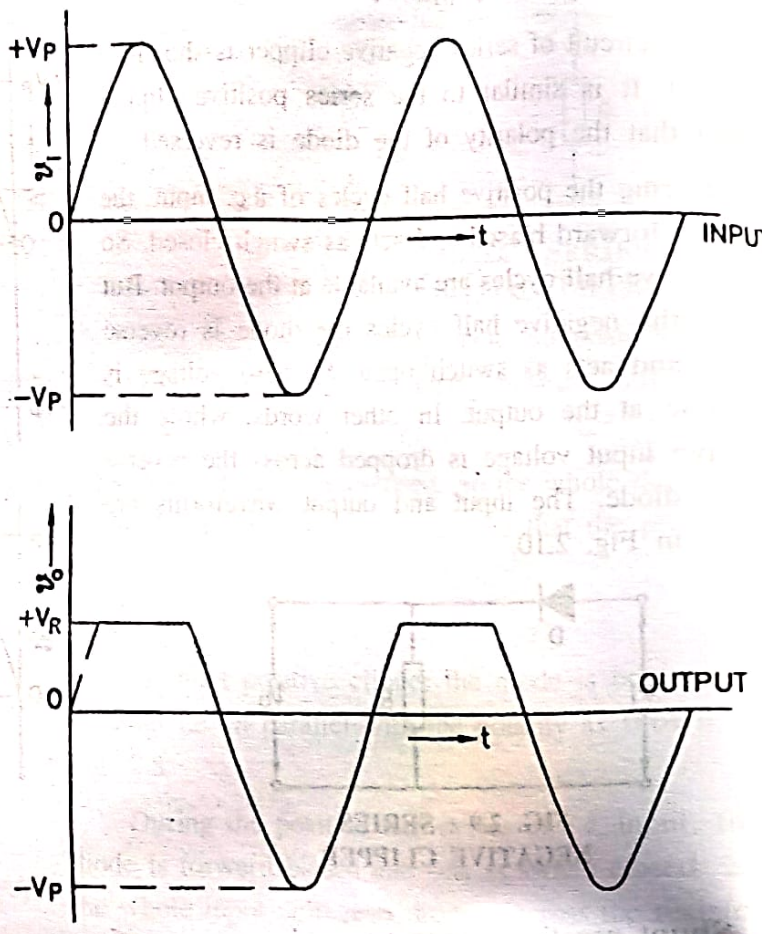


FIG. 2.13 : WAVEFONNS OF SERIES AND SHUNT TYPE BIASED POSITIVE CLIPPER

When the input voltage ϑ_i is more than V_R, the diode is reverse biased and acts as a switch open. the output voltage is V_R.

Thus in this the whole positive cycle is not clipped off but only peak portion of the positive half cyc above V_R is clipped off. By changing the battery emf, V_R can be varied and the desired portion can be clipp off. The waveforms are shown in Fig. 2.13.

(2) Shunt type biased positive clipper :

The circuit is shown in Fig. 2.14. Diode is connected in shunt. Battery of emf V_R is connected in series with the diode with the polarity as shown.

The diode conducts as long as the input voltage is greater than $+V_R$. It works as switch closed. So the output remains at V_R. When the input voltage is less than $+V_R$, the diode is reverse biased and acts as switch open. So the input voltage appears at the output terminals. This includes the negative half cycle as it is less than $+V_R$, Waveforms are shown in Fig. 2.13.

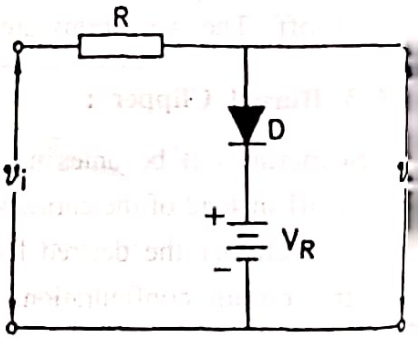


FIG. 2.14 : SHUNT TYPE BIASED POSITIVE CLIPPER

Biased positive clipper with the polarity of the battery reversed :

Let us consider the situations when the polarity of the battery is reversed in each of the above cases

(i) Series type biased positive clipper with the reverse polarity of the battery :

The circuit is shown in Fig. 2.15. It is seen that the polarity of the battery V_R is reversed.

When the input voltage rises from zero and goes to positive peak the diode is reverse biased so $-V_R$ is available at the output. This continues until the input voltage becomes less than $-V_R$. During the period when the input voltage is less than $-V_R$, the diode remains forward biased and acts as switch closed. So the input voltage is available at the output terminals. It continues upto the point when input voltage reaches to $-V_R$. When the input voltage rises from $-V_R$ to zero, the diode remains reverse biased and again $-V_R$ is available across the output. Thus signal above $-V_R$ is clipped off. Waveforms are shown in Fig. 2.16.

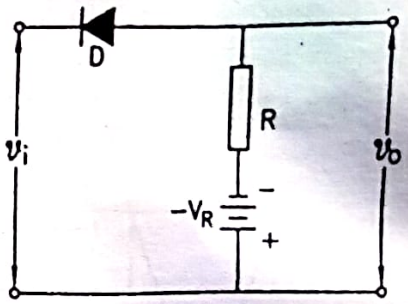


FIG. 2.15 : SERIES TYPE BIASED POSITIVE CLIPPER WITH THE REVERSE POLARITY OF THE BATTERY REVERSED

(ii) Shunt type biased positive clipper with the reverse polarity of the battery :

The circuit is shown in Fig. 2.17.

During the positive half cycles of a.c. input, the diode is forward biased, and acts as a closed switch. So the output remains $-V_R$. When the input voltage decreases to zero and goes to negative and becomes less than $-V_R$, the diode is reverse biased. So the input waveform appears at the output. Again when the input voltage increases from $-V_R$ to zero, the output becomes $-V_R$ as the diode is forward biased. Thus any signal voltage above $-V_R$ is clipped off. The waveforms are shown in Fig. 2.16.

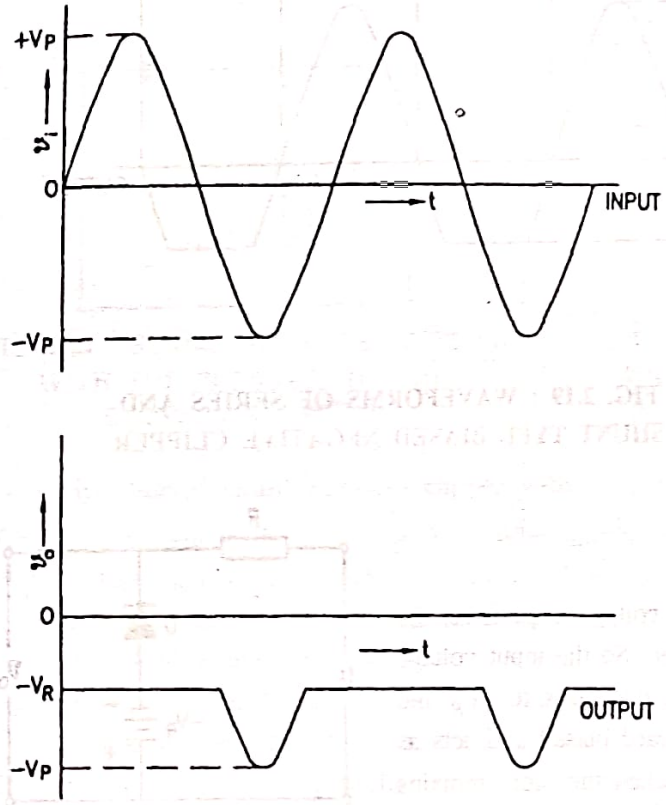


FIG. 2.16 : WAVEFORMS OF SERIES AND SHUNT TYPE BIASED POSITIVE CLIPPER WITH THE REVERSE POLARITY OF THE BATTERY

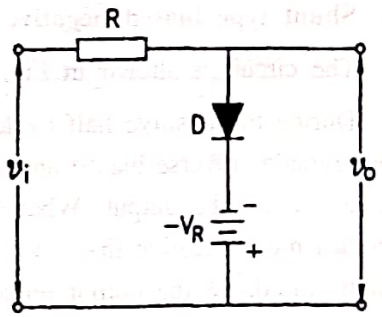


FIG. 2.17 : SHUNT TYPE BIASED POSITIVE CLIPPER WITH THE REVERSE POLARITY OF THE BATTERY

(b) Biased negative clipper :

In biased negative clipper also a battery of emf V_R is connected. In this instead of clipping off the whole negative half of the input wave, only some portion near the peak is clipped off. This depends upon the value of V_R. In this also two versions are available i.e. series type and shunt type.

(1) Series type biased negative clipper :

Circuit of a series type biased negative clipper is shown in Fig. 2.18,

Diode remains forward biased and acts as switch closed until the input voltage becomes less than $-V_R$ so input voltage is available at the output. When the input voltage becomes less than $-V_R$, the diode is reverse biased and acts as switch open. So $-V_R$ voltage is available until the input voltage becomes less negative than $-V_R$. The waveforms are shown in Fig. 2.19.

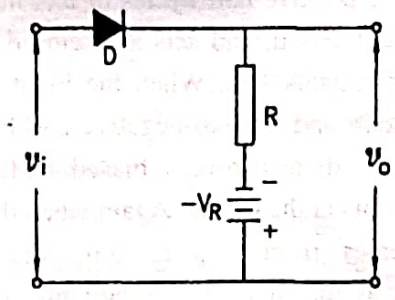


FIG. 2.18 : SERIES TYPE BIASED NEGATIVE CLIPPER

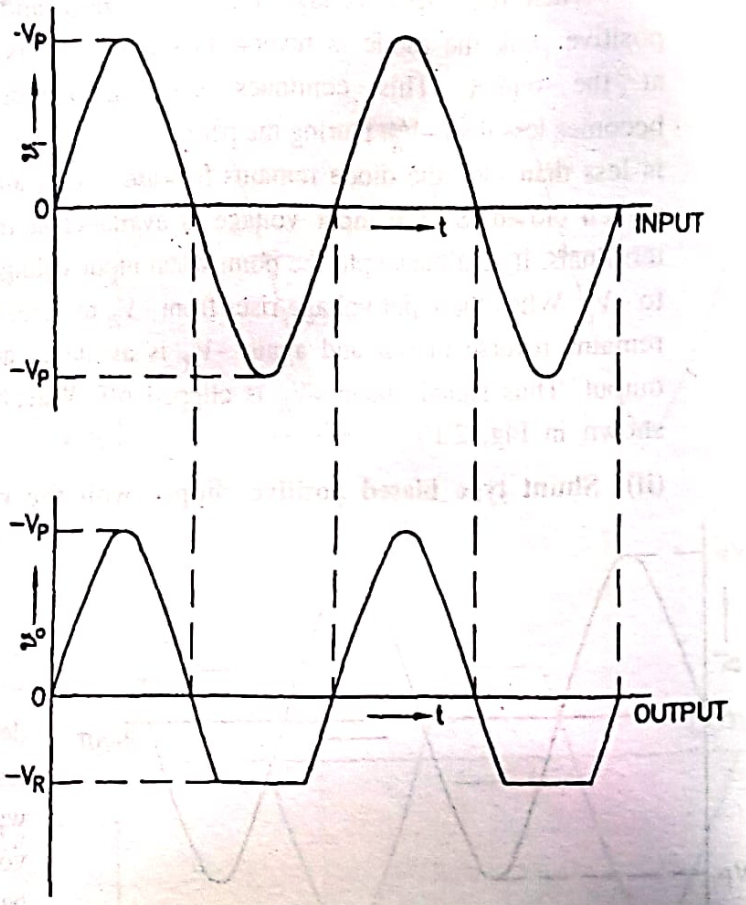


FIG. 2.19 : WAVEFORMS OF SERIES AND SHUNT TYPE BIASED NEGATIVE CLIPPER

(2) Shunt type biased negative clipper :

The circuit is shown in Fig. 2.20.

During the positive half cycle when the input voltage is positive, the diode remains reverse biased and acts as switch open. So the input voltage appears across the output. When the input voltage decreases to $-V_R$ and becomes more negative than $-V_R$, the diode is forward biased and acts as switch closed. So the output remains $-V_R$. Again when the input remains less negative the diode becomes reverse biased and the input appears at the output. The waveforms are shown in Fig. 2.19.

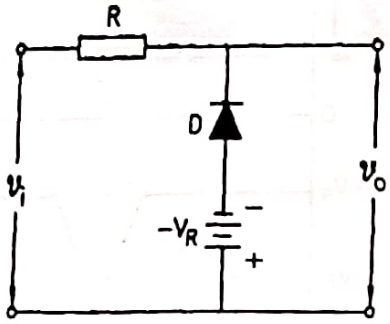


FIG. 2.20 : SHUNT TYPE BIASED NEGATIVE CLIPPER

Biased negative clipper with the polarity of the battery reversed :

In this also there are two types :

(i) Series type and

(ii) shunt type.

(i) Biased series negative clipper with the polarity or the battery reversed :

The circuit is shown in Fig. 2.21. In this the polarity of the battery is reversed.

The diode remains reverse biased and acts as switch open until the input voltage reaches to $+V_R$. So output remains +VR. When the input voltage becomes more than $+V_R$, the diode is forward biased and the input is available at the output. Again when the input becomes less than $+V_R$, the output remains to $+V_R$. The waveforms are shown in Fig. 2.22.

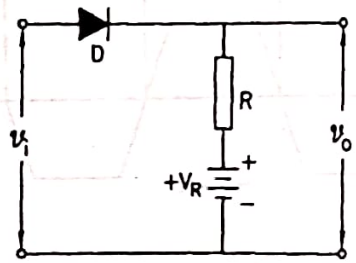


FIG. 2.21 : SERIES NEGATIVE CLIPPER WITH THE POLARITY OF THE BATTERY REVERSED

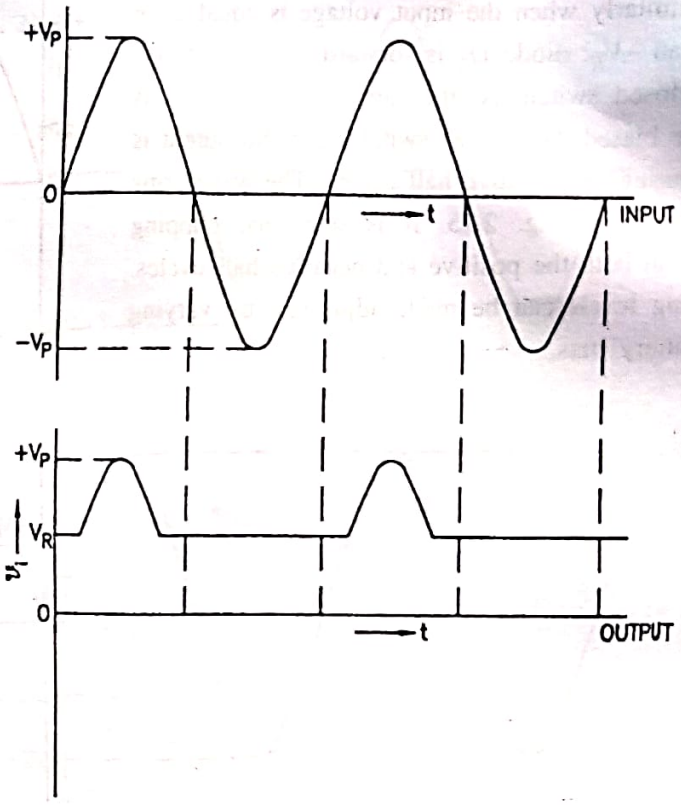


FIG. 2.22 : WAVEFORMS OF SERIES AND SHUNT NEGATIVE BIASED CLIPPER WITH POLARITY OF THE BATTERY REVERSED

(ii) Biased shunt negative clipper with polarity of the battery reversed :

Circuit is shown in Fig. 2.23. When the input voltage is less than $+V_R$, the diode is forward biased and acts as switch closed. So $+V_R$ is available at the output. During the period when the input voltage is more than $+V_R$, the diode is reverse biased and acts as switch open. So the input voltage is available at the output terminals. So any signal voltage less than $+V_R$ is clipped off. Waveforms are shown in Fig. 2.22.

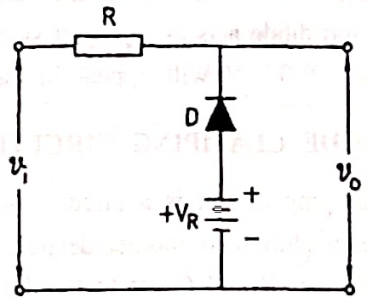


FIG. 2.23 : BIASED SHUNT NEGATIVE CLIPPER WITH POLARITY OF THE BATTERY REVERSED

2.1.6.4 Combined Clipper :

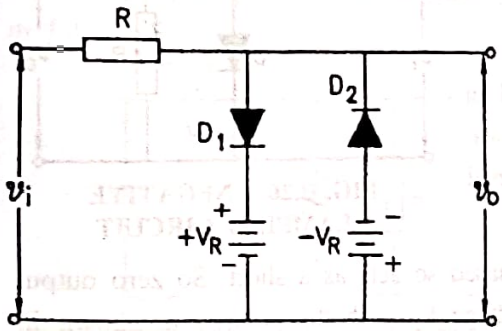


FIG. 2.24 : COMBINED CLIPPER

With the help of the combined clipper, both the positive and negative clipping can be achieved simultaneously. Circuit is shown in Fig. 2.24. It is seen that two shunt type biased clipper circuits are connected together. One is shunt positive biased clipper and other is shunt negative biased clipper.

When the input voltage is more than or equal to V_R, diode D_1 is forward biased and acts as switch closed at the same time diode D_2 is reverse biased and it acts as switch open. So the output voltage is limited to $+V_R$ during the positive half cycles.

Similarly when the input voltage is equal to or less than $-V_R$, diode D_2 is forward biased and acts as a closed switch. At the same time diode D_1 is reverse biased. So acts as switch open. So output is $-V_R$ during the negative half cycles. The waveforms are shown in Fig. 2.25. It is seen that clipping occurs in both the positive and negative half cycles. Clipping levels can be made adjustable by varying the battery bias.

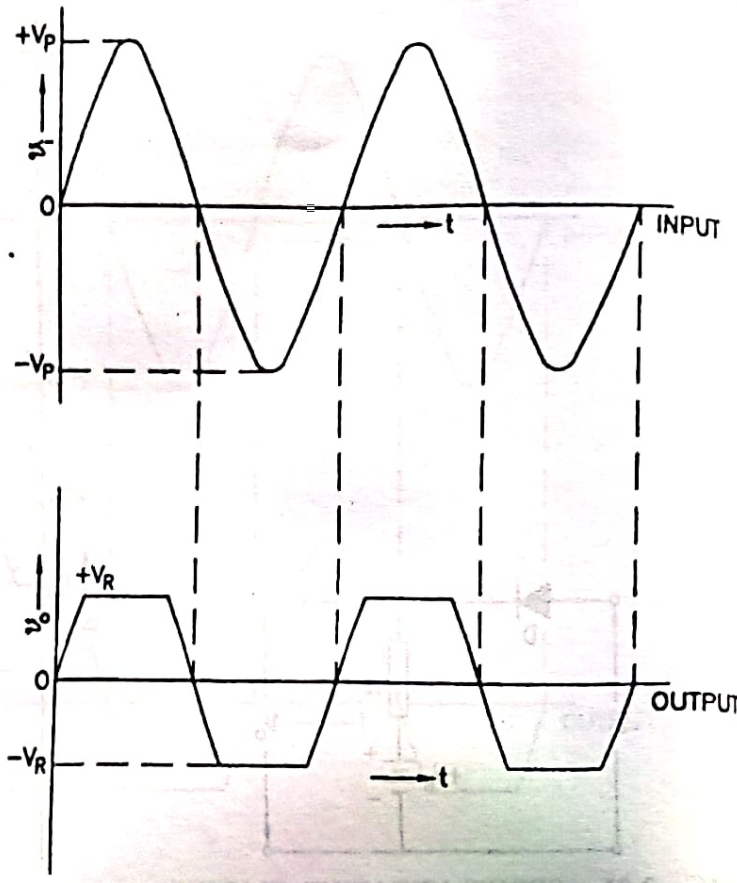


FIG. 2.25 : WAVEFORMS OF THE COMBINED CLIPPI

Applications of clipping circuits :

Clipping circuits are used in radio receiver, T.V. receiver, radar, digital and other electronic circuits.

Note : In all the circuits the first approximation for the diode is used i.e. the voltage drop across the dio is zero and diode acts as a perfect switch. If second approximation is used in which the voltage drop is considere an offset of 0.7 V will appear in the output waveforms accordingly.

2.2 DIODE CLAMPING CIRCUITS :

Clamping circuit is a circuit with the help of which the d.c. level is introduced to an a.c. signal. That the level is shifted to another desired d.c. level. The clamping circuit needs a diode, a capacitor C and a resisto The values of R and C are to be chosen such that the time constant $T = RC$ is sufficiently high. There are tw types of clamping circuits :

(i) Negative Clamping Circuit and

(ii) Positive Clamping Circuit.

2.2.1 Negative Clamping Circuit :

In negative clamping circuit the d.c. level of the signal is shifted in negative direction. Diode D, resistor R and capacitor C are connected as shown. The Input may be sine wave, square wave or any other wave whose level has to be shifted. Circuit is shown in Fig. 2.26.

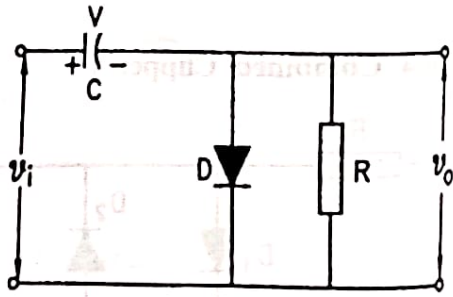


FIG. 2.26 : NEGATIVE CLAMPING CIRCUIT

During the positive half cycles of the input, the diode is forward biased so acts as a short. So zero output voltage is available. During this interval, the capacitor C charges to voltage V_P with the polarity as shown in the Fig. 2.26 i.e. the left plate positive and the right plate negative.

During the negative half cycle, the diode is reverse biased. So acts as an open switch. Output is developed across the resistor. The voltage across the resistor is the supply voltage plus the voltage across the capacitor. i.e.

$$V_0 = -V_P + (-V_P) = -2-V_P$$

Waveforms are shown in Fig. 2.27.

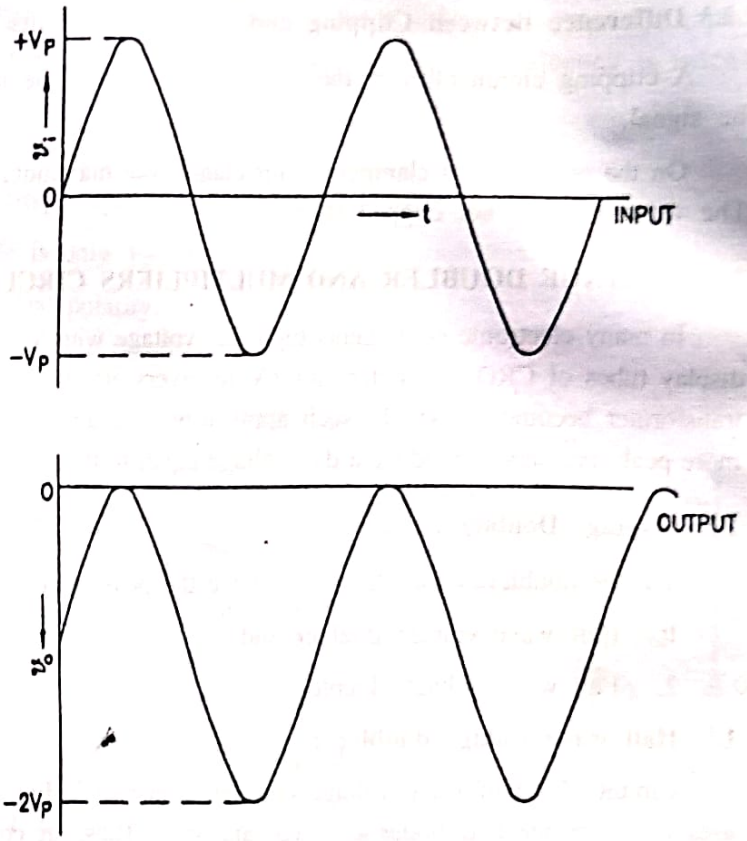


FIG. 2.27 : WAVEFONNS OF THE NEGATIVE CLAMPER

2.2.2 Positive Clamping Circuit :

The circuit of a positive clamper is shown in Fig. 2.28. It is seen that the polarity of the diode is reversed from that shown in Fig. 2.26.

During the positive half cycle, the diode is reverse biased and acts as a switch open. When the input signal becomes negative the diode is forward biased and acts as switch closed. The charging current flows and the capacitor charges to input voltage V with the polarity as shown i.e. the left plate negative and the right plate positive. Again during the positive half cycle of the input, the diode is reverse biased and acts a switch open. So the output voltage across the resistor is $V_P + V_P = 2V_P$. Waveforms are shown in Fig. 2.29.

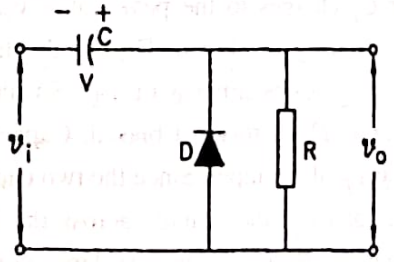


FIG. 2.28 : POSITIVE CLAMPING CIRCUIT

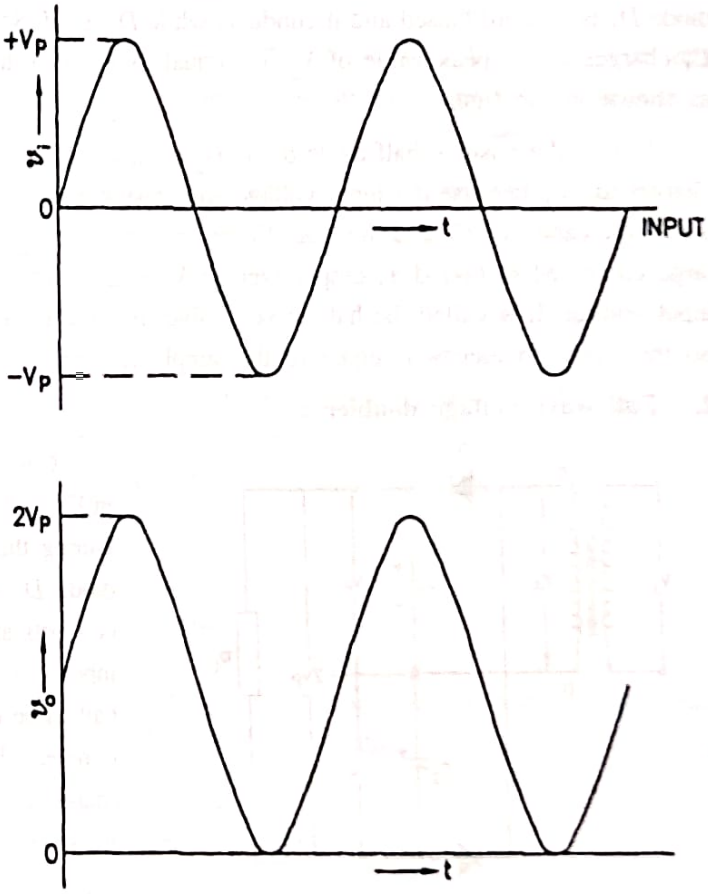


FIG. 2.29 : WAVEFORMS OF POSITIVE CLAMPER

2.2.3 Difference Between Clipping and Clamping Circuits :

A clipping circuit clips of the unwanted portion of the input waveform. It does not shift the d.c. level of the signal.

On the other hand a clamper circuit clamps the maximum positive or negative peak to the desired d.c. level. The wave shape is not clipped off.

2.3 VOLTAGE DOUBLER AND MULTIPLIERS CIRCUITS :

In many electronic equipments high d.c. voltage with low current is required for their operation. For example display tubes of CRO, computer and TV receivers etc. If this high voltage is obtained by using transformer, the transformer becomes bulky. In such applications voltage multipliers are used. A voltage multiplier uses two or more peak rectifiers to produce a d.c. voltage equal to the multiple of the peak input voltage i.e. $2V_P$, $3V_P$, $4V_P$ etc.

2.3.1 Voltage Doubler :

Voltage doubler is used to obtain twice the peak input voltage. There are two types of the voltage doubler

1. Half wave voltage doubler and
2. Full wave voltage doubler

1. Half wave voltage doubler :

Circuit of a half wave voltage doubler is shown in Fig. 2.30. It uses a transformer, two diodes and two capacitors. These are connected as shown.

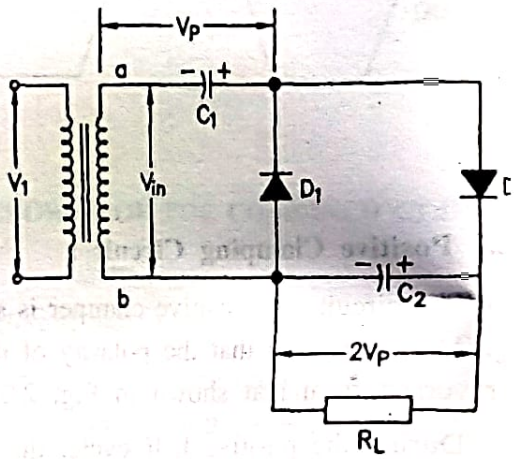


FIG. 2.30 : HALF WAVE TYPE VOLTAGE DOUBLER

During the negative half cycle of V_{in}, 'a' is negative w.r.t. 'b', diode D_1 is forward biased and it conducts while D_2 is off. So capacitor C_1 charges to the peak value of V_{in} i.e. equal to V_P with the polarity as shown in the figure.

During the positive half cycle diode D_2 conducts and capacitor C_2 charges to $2V_P$ because the input voltage and capacitor C_1 are in series across the capacitor C_2 and the load. Since time constant $R_L C_2$ is very large compared to period T, output voltage $V_0 = 2V_P$. Thus it is seen that the output voltage is twice the peak input voltage. It is called the half wave doubler because capacitor C charges only during the negative half cycles. So the ripple frequency is equal to the supply frequency.

2. Full wave voltage doubler :

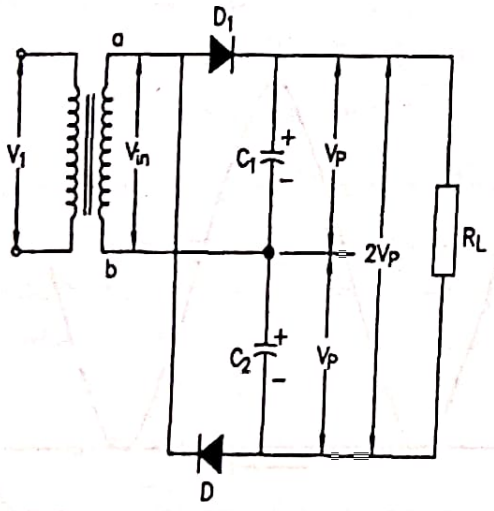


FIG. 2.31 : FULLWAVE VOLTAGE DOUBLER

Circuit diagram of a full wave type voltage doubler is shown in Fig. 2.31. In this also two diodes and two capacitors are used. During the positive half cycle of a.c. 'a' is positive w.r.t. 'b', diode D_1 is forward and D_2 is reverse biased. So diode D_1 conducts and capacitor C_1 charges to the peak value V_P of the input voltage with the, polarity as shown. During the negative half cycle terminal 'b' becomes positive w.r.t. 'a'. So diode D_1 is reverse biased and diode D_2 is forward biased. Capacitor C_2 charges to the peak value V_P of the input. Since the two capacitors are in series across the load R_L, the voltage across the load is $2V_P$. Since the time constant $R_L C_1 C_2$ is very large compared to the time period T of the input a.c., $V_0 = 2V_P$.

It is called the full wave doubler because capacitor C_1 is charged during the positive half cycles and capacitor C_2 charges during the negative half cycles i.e. both the half cycles are utilized. Ripple frequency is twice the supply frequency.

Advantages :

1. Ripple frequency is twice the supply frequency so it is easy to filter the output wave.
2. Peak inverse voltage across each diode is only V_P.
3. When the input is grounded, it gives dual polarity.

Disadvantages :

There is no common ground available between the input and output.

2.3.2 Voltage Tripler :

Circuit of voltage tripler is shown in Fig. 2.32. It is constructed by adding one peak rectifier to the half wave voltage doubler.

During negative half cycles capacitor C_1 charges to V_P volts and capacitor C_3 charges to $2V_P$ with the polarity as shown. During the positive half cycles capacitor C_2 charges to $2V_P$. Output is taken as shown. It is seen that the Output voltage is the sum of the voltages across C_1 and C_3 in series i.e. $3V_P$. Since the time constant R_L C_1 C_3 is much greater than the time period T, the output voltage $V_0 = 3V_P$.

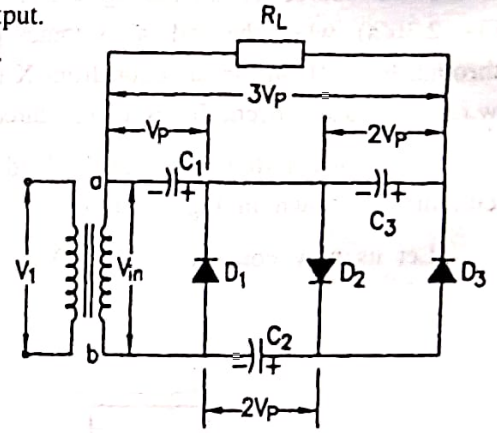


FIG. 2.32 : VOLTAE TRIPLER

2.3.3 Voltage Quadrupler :

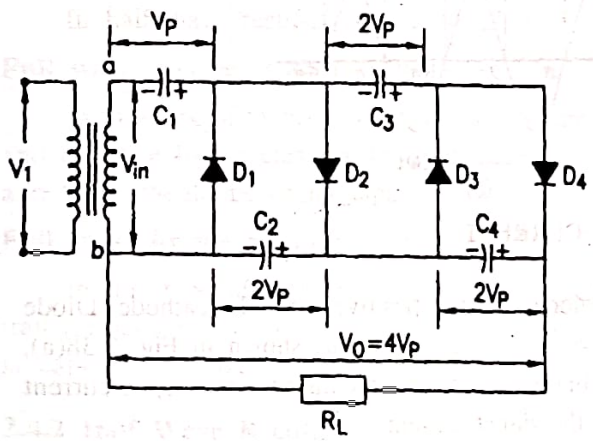


FIG. 2.33 : VOLTAGE QUADRUPLER

Voltage quadrupler is obtained by adding one more section of the peak rectifier to the tripler. The circuit is shown in Fig. 2.33.

Capacitor C_1 charges to V_P volts while capacitors C_2, C_3 and C_4 are each charged to $2V_P$ volts. Output voltage is taken as shown. Voltage output is the sum of the voltage across capacitors C_2 and C_4 i.e. $4V_P$ volts. Time Constant R_L C_2 C_4 is much higher than the time period T.

If higher voltage output is desired say $5V_P$, $6V_P$, $7V_P$ etc. then the more numbers of required sections of the peak rectifiers can be added and the output voltage is taken accordingly.

2.4 PN JUNCTION DIODE AS RECTIFIER :

Rectifier :

A rectifier is a circuit by which the alternating current is converted into the direct current and the rectification is the process by which the alternating current is converted into the direct current.

FIG. 2.34 : BLOCK DIAGRAM OF RECTIFIER

Junction diode as rectifier :

When PN junction diode is forward biased, it offers very low resistance and allows the current to flow easily Fig. 2.35(a). But when it is reverse biased Fig. 2.35(b), it offers very high resistance and negligible reverse current flows. In rectifier use is made of this property of the PN junction diode.

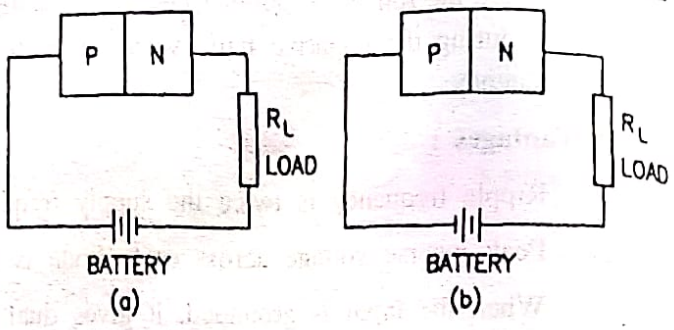


FIG. 2.35 : JUNCTION DIODE (a) FORWARD BIASED AND (b) REVERSE BIASED

In Fig. 2.36(a), a load resistance R_L is connected across the source of alternating current. As shown in Fig. 2.36(a), when the end 'a' becomes positive w.r.t. the end 'b', in positive half cycle, electric current flows through load RL in the direction from X to Y. And in the negative half cycle when terminal 'b' becomes positive w.r.t. 'a', load current flows in the direction from Y to X.

Thus current flows through the load alternately in both the directions. Waveforms of input voltage and load current are shown in Fig. 2.36(c).

Let us now connect a diode D in the circuit as shown in Fig. 2.37(a).

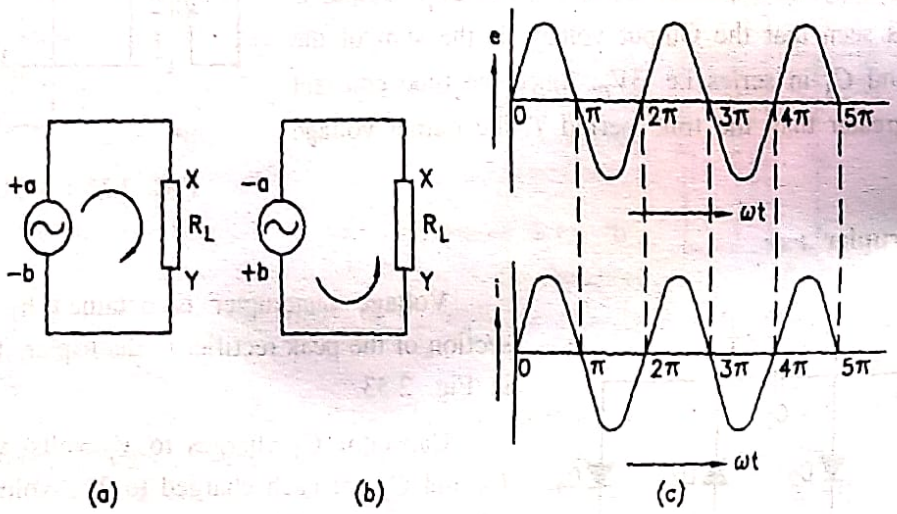


FIG. 2.36 : ALTERNATING CURRENT

When terminal 'a' becomes positive w.r.t. 'b', anode of the diode becomes positive w.r.t. the cathode. Diode is forward biased and allows the current to flow, so current flows from X to Y. But as shown in Fig. 2.38(a), when terminal 'a' becomes negative w.r.t. 'b', the diode is reverse biased and offers very high resistance, so current can not flow. This is how alternating current is converted into the direct current.

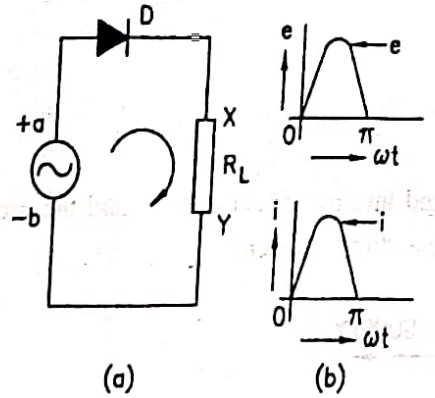


FIG. 2.37(a) : CONNECTION OF DIODE IN CIRCUIT (b) WAVEFORMS

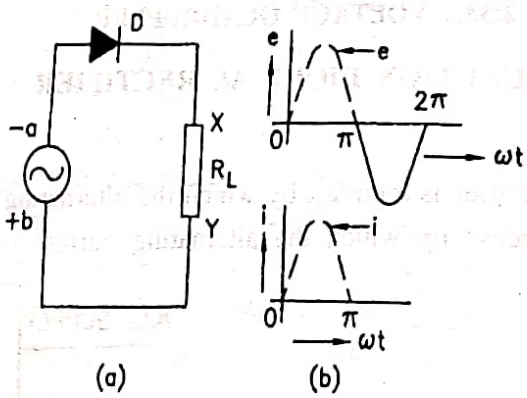


FIG. 2.38(a) : CURRENT DOES NOT FLOW (b) WAVEFORM IN NEGATIVE HALF CYCLE

2.4.1 Classification of Single-Phase Rectifier :

There are two main types of electronic rectifier.

1. Single-phase rectifier and
2. Poly phase rectifier.

There are two types of single-phase rectifier.

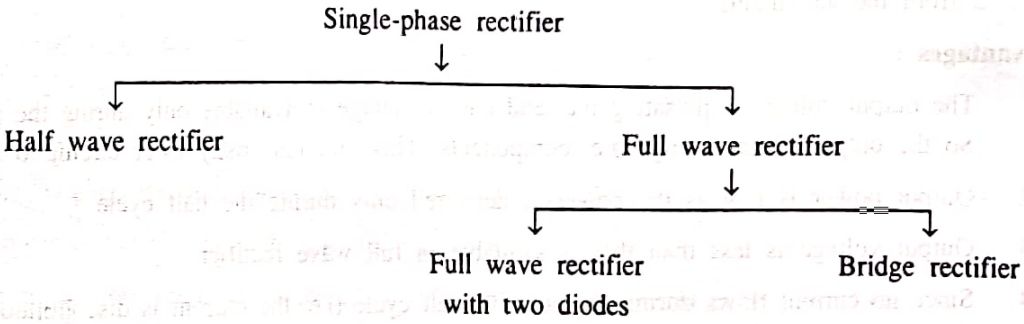
1. Half wave rectifier and
2. Full wave rectifier.

There are two types of single-phase full wave rectifier

(a) Full wave rectifier with two diodes and

(b) Bridge rectifier.

We can show this classification as follows.



Half wave rectifier :

In half wave rectifier conduction of current takes place only during the positive or negative half cycles.

Full wave rectifier with two diodes :

In this type of rectifier circuit the current through the load flows in the same direction during both the positive and negative half cycles, so this type of rectifier is known as the full wave rectifier. Two diodes are required and there should be center taping in the secondary of the transformer.

Full wave bridge rectifier :

In this type of rectifier circuit, it is not necessary to have centre taping in the secondary winding of the transformer but four diodes are required. As the circuit resembles to the Wheatstone bridge, this rectifier circuit is called the bridge rectifier.

2.4.2 Half Wave Rectifier :

In half wave rectifier, current flows only during the half cycle of a.c.

Circuit :

Half wave rectifier circuit is shown in Fig. 2.39(a), and the waveforms of the input and output are shown in Fig. 2.39(b).

Diode D and load resistance are connected in series across the secondary winding of a step down transformer (it may be step up type also).

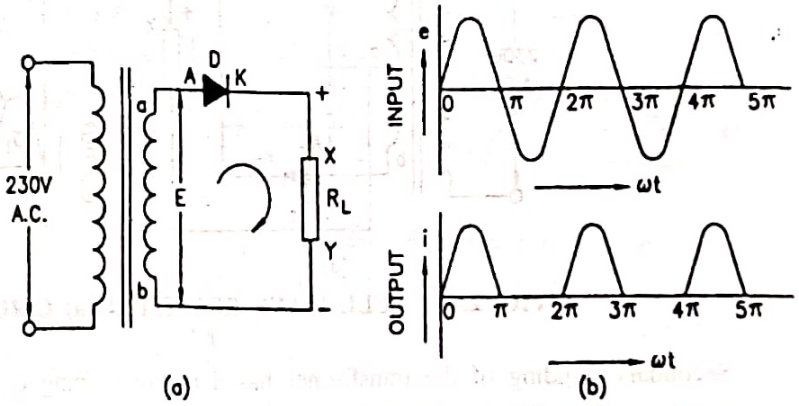


FIG. 2.39 : HALF WAVE RECTIFIER (a) CIRCUIT (b) WAVEFORMS

Working :

When terminal 'a' becomes positive w.r.t. 'b' in the positive half cycle of a.c., anode A of the diode D becomes positive w.r.t. its cathode K, the diode is forward biased and offers very low resistance. Electric current flows through the load in the direction a A K X Y b a. That is the current flows from X to Y through the load. But during the negative half cycle when terminal 'a' becomes negative w.r.t. 'b', the diode is reverse biased and offers very high resistance and current does not flow. Waveforms of the input voltage and output current are shown in Fig. 2.39(b).

It is seen from the waveforms that current conducts only in one direction. That is current flows during half cycle of a.c. So it is called the halfwave rectifier.

Function of transformer :

Voltage can be stepped up or down with the help of a transformer. In rectifier definite a.c. voltage has to be applied to the rectifier in order to get the definite d.c. voltage output. Now single phase a.c. voltage is available at 230 V. So using transformer necessary a.c. voltage is obtained. Moreover when transformer is used, the circuit is isolated from the a.c. mains.

Disadvantages :

1. The output voltage is pulsating d.c. and output voltage is available only during the positive half cycle. So the output contains large a.c. components. This requires costly filter circuit to obtain smooth d.c.
2. Output power is low as the power is delivered only during the half cycle.
3. Output voltage is less than that is available in full wave rectifier.
4. Since no current flows during the negative half cycle (i.e. the current is discontinuous) the choke type filter can not be used.

2.4.3 Wave Rectifier with Two Diodes :

In this type of rectifier, both the positive and negative half cycles of a.c. are converted into d.c. i.e. the d.c. load current flows during both the half cycles so it is called the full wave rectifier. This requires two diodes and it is necessary to have centre tapping in the secondary winding of the transformer.

Circuit :

The circuit of a full wave rectifier using two diodes is shown in Fig. 2.40(a).

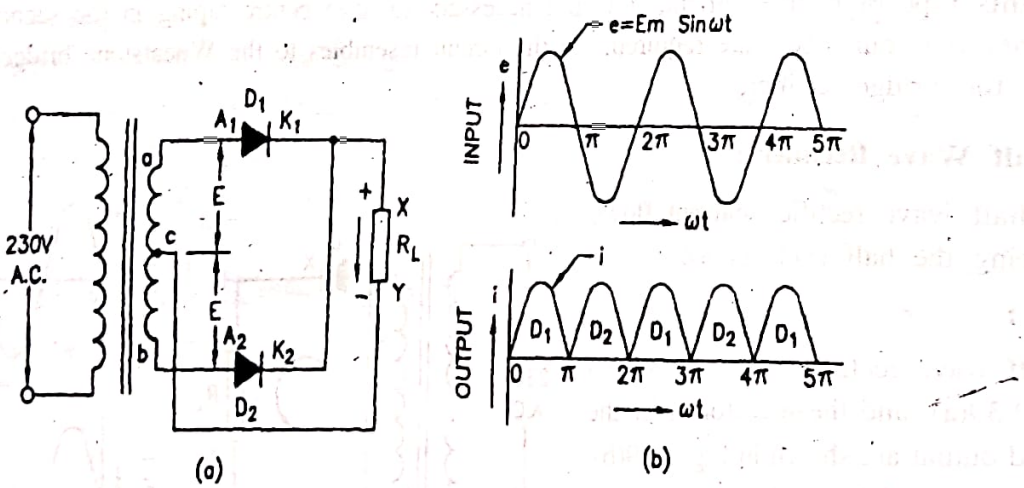


FIG. 2.40 : FULL WAVE RECTIFIER (a) CIRCUIT (b) WAVEFORMS

Secondary winding of the transformer has a centre tapping c. D_1 and D_2 are two diodes whose cathodes are joined together. This is the positive terminal of the d.c. Anodes are connected to the two outer terminals

of the transformer secondary. Centre tapping becomes the negative terminal of the d.c. output. Load is connected across the positive and negative terminals.

Working :

When terminal 'a' becomes positive w.r.t. 'c' during the positive half cycle of a.c., at the same time terminal 'b' becomes negative w.r.t. 'c'. So diode D_1 becomes forward biased and diode D_2 becomes reverse biased. So diode D_1 conducts and allows the current to conduct, where – as diode D_2 does not allow the current to flow. So electric current flows in the direction a A_1 K_1 X Y c a.

During the negative half cycle when terminal 'a' becomes negative w.r.t. 'c', terminal 'b' becomes positive. Diode D_1 is reverse biased and diode D_2 is forward biased. So D_1 cannot conduct but D_2 does. Electric current flows in the direction b A_2 K_2 X Y c b. During this time also the load current flows from X to Y. Terminal X becomes positive and Y becomes negative of the d.c.

It Fig. 2.40(b) are shown the waveforms. Output waves do not enter in the negative direction, and current flows during both the half cycles.

In this type of rectifier two diodes are used and it is necessary to have centre tapping in the secondary winding of the transformer. Value of d.c. voltage is double than that is available in the half wave rectifier.

Advantages :

1. Output d.c. voltage is twice than that with half wave rectifier.
2. Output power is twice than with half wave rectifier.
3. Only two diodes are required compared to the four required in bridge type rectifier.

Disadvantages :

1. It is necessary to have centre tapping in the secondary winding of the transformer.
2. Peak inverse voltage across the diode is more so diodes with higher PIV rating are to be used than with diodes used in full wave bridge rectifier.
3. Secondary utilization of the transformer is less as when half of tile winding is working the remaining half winding is not utilized.

2.4.4 Full Wave Bridge Rectifier :

In this type of rectifier there is no need to have centre tapping in the secondary winding of the transformer but four diodes are required in place of two. The circuit configuration is similar to the Wheatstone bridge, so the circuit is called the bridge rectifier.

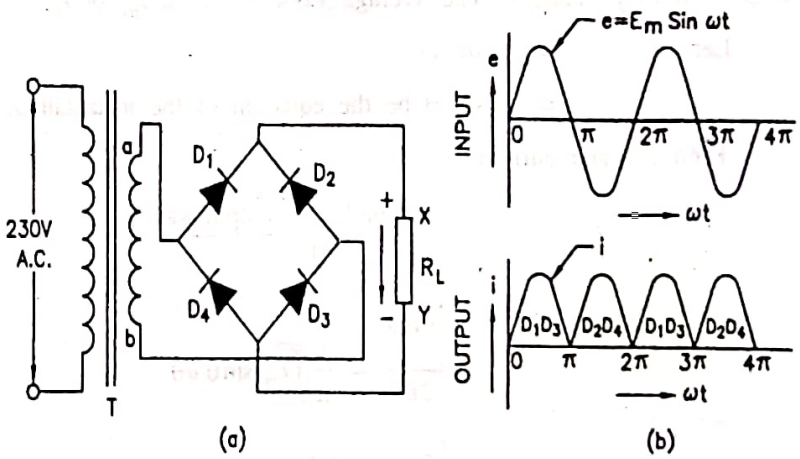


FIG. 2.41 : FULL WAVE BRIDGE RECTIFIER
(a) CIRCUIT (b) WAVEFORMS

Circuit :

T is a transformer. Cathodes of diodes D_1 and D_2 are joined together which becomes the positive terminal of d.c. Anodes of diodes D_3 and P_4 are joined together which becomes the negative terminal of the d.c. Anode c of D_1 and cathode of D_4 are joined together and is connected to terminal 'a' of the transformer. Similarly cathode of D_3 and anode of D_2 are joined together and connected to the terminal 'b' of the transformer. Load is connected across the positive and negative terminals of the d.c. output.

Working :

During the positive half cycle of a.c., when terminal 'a' becomes positive w.r.t. 'b', diodes D_1 and D_3 are forward biased and allow the current to conduct. At the same time diodes D_2 and D_4 are reverse biased and can not conduct.

Current flows in the direction a D_1 X Y D_3 b a (Fig. 2.42(a)).

During the negative half cycle terminal 'b' becomes positive w.r. to 'a'. Diodes D_2 and D_4 are forward biased and D_1 and D_3 are reverse biased. So diodes D_2 and D_4 conduct and D_1 and D_3 can not. The current flows in the direction b D_2 X Y D_4 a b as shown in Fig. 2.42(b).

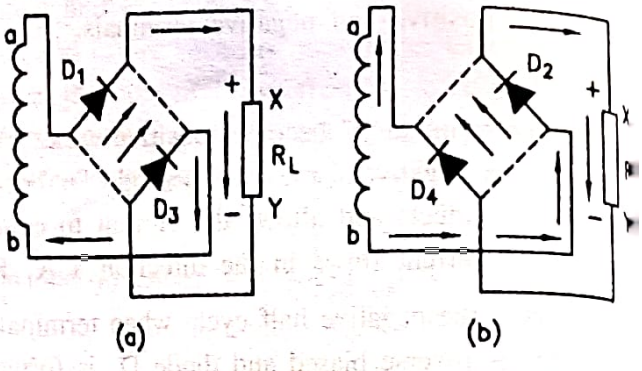


FIG. 2.42 : CONDUCTION OF CURRENT (a) DURING POSITIVE HALF CYCLE AND (b) DURING NEGATIVE HALF CYCLE

It is seen that the load current flows from X to Y in both the half cycles. Waveforms are shown Fig. 2.42(b).

Advantages :

1. Simple transformer without the centre tapping in secondary is needed.
2. Peak inverse voltage across the diode is half than that in the full wave rectifier using two diod
3. For the same secondary voltage, the output d.c. voltage is twice than that in the full wave rectifier two diodes.

Disadvantages :

1. Four diodes are required.
2. Two diodes conduct in series so the voltage drop in the diode is twice. This becomes important the output voltage is low.

2.5 VALUES OF I_{dc}, E_{dc} AND I_{RMS} FOR HALF WAVE RECTIFIER :

Value of the output current of the rectifier does not remain the same, but it continuously changes. The average value is called I_{av} or I_{dc}.

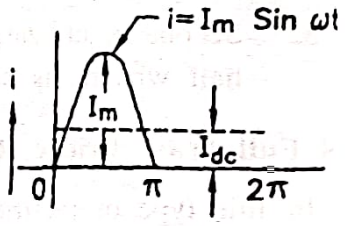


FIG. 2.43 : OUTPUT WA OF HALF WAVE RECTIFI

Let $i = I_m \sin \omega t$

$= I_m \sin \theta$ be the equation of the instantaneous current.

Then average current

$$I_{av} = I_{dc} = \frac{\text{Area under the output wave}}{\text{base}} \quad \ldots$$

$$I_{dc} = \frac{\int_0^{\pi} i \, d\theta}{2\pi} = \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{2\pi}[-\cos 0]_0^{\pi}$$

$$= \frac{-I_m}{2\pi}[\cos\pi - \cos 0]$$

$$\boxed{I_{dc} = \frac{I_m}{\pi}} \quad \ldots$$

Now $E_{dc} = I_{dc} \times R_L$ and $I_m = \frac{E_m}{R_L}$

$$E_{dc} = \frac{I_m}{\pi} \times R_L$$

$$= \frac{E_m}{\pi R_L} \times R_L$$

$$\boxed{E_{dc} = \frac{E_m}{\pi}} \quad \text{... (3)}$$

and $$I_{RMS} = \sqrt{\frac{\int_0^{\pi} i^2 \, d\theta}{2\pi}} = \sqrt{\frac{1}{2\pi} \int_0^{\pi} I_m^2 \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \cdot \int_0^{\pi} \frac{1 - \cos 2\theta}{2} \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \left[\int_0^{\pi} d\theta - \int_0^{\pi} \cos 2\theta \, d\theta \right]}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \left[\theta \Big|_0^{\pi} - \frac{\sin 2\theta}{2} \Big|_0^{\pi} \right]}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \left[(\pi - 0) - \frac{1}{2} \{\sin 2\pi - \sin 0\} \right]}$$

$$\boxed{I_{RMS} = \frac{I_m}{2}} \quad \text{... (4)}$$

2.6 VALUES OF I_{dc}, E_{dc} AND I_{RMS} FOR FULL WAVE RECTIFIER :

Output waveform of full wave rectifier is shown in Fig. 2.44.

$$I_{dc} = \frac{\text{Area under the output wave}}{2\pi}$$

$$= \frac{2 \int_0^{\pi} i \, d\theta}{2\pi} = \frac{1}{\pi} \left[\int_0^{\pi} I_m \sin\theta \, d\theta \right]$$

$$= \frac{I_m}{2\pi} [-\cos 0]_0^{\pi}$$

$$= \frac{-I_m}{2\pi} [\cos\pi - \cos 0]$$

$$= \frac{-I_m}{\pi} [-1 - 1]$$

$$\boxed{I_{dc} = \frac{2I_m}{\pi}} \quad \text{... (5)}$$

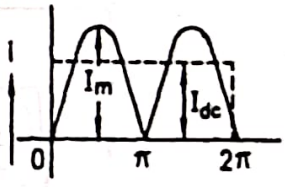


FIG. 2.44 : OUTPUT WAVE OF FULL WAVE RECTIFIER

and $$E_{dc} = I_{dc} \times R_L$$

$$= \frac{2I_m}{\pi} \times R_L$$

$$= \frac{\pi}{2} \frac{E_m}{R_L} \times R_L$$

$$\boxed{E_{dc} = \frac{2E_m}{\pi}} \quad \ldots (6)$$

and $$I_{RMS} = \sqrt{\frac{2}{2\pi} \int_0^{\pi} i^2 \, d\theta}$$

$$= \sqrt{\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{\pi} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \left[\int_0^{\pi} d\theta - \int_0^{\pi} \cos 2\theta \, d\theta \right]}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \left[\theta \Big|_0^{\pi} - \frac{\sin 2\theta}{2} \Big|_0^{\pi} \right]}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \left[(\pi - 0) - \frac{1}{2} \{\sin 2\pi - \sin 0\} \right]}$$

$$\boxed{I_{RMS} = \frac{I_m}{\sqrt{2}}} \quad \ldots (7)$$

2.7 RIPPLE FACTOR :

It is seen that the output d.c. available is not pure d.c. but it is the mixture of many waves of a.c. of different frequencies. The percentage of this a.c. components in the output can be found from the ripple factor. Ripple factor is denoted by γ. It is defined as the ratio of RMS value of a.c. components in the output to the d.c. component.

$$\gamma = \frac{\text{RMS value of a.c. components in the output}}{\text{d.c. component}} \quad \ldots (8)$$

$$\gamma = \sqrt{\left(\frac{I_{RMS}}{I_{dc}} \right)^2 - 1} \quad \ldots (9)$$

Let us now derive this relation.

Let i be the instantaneous value of the output load current. This is the sum of direct current I_{dc} and the a.c. component i'.

$$i' = I_{dc} + i' \quad \ldots (10)$$

$$i' = i - I_{dc} \quad \ldots (11)$$

RMS value of a.c. component in the output

$$I'_{RMS} = \sqrt{\frac{1}{\pi}\int_0^{2\pi}(i')^2\,d\theta} \quad \text{... (12)}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi}(i - I_{dc})^2\,d\theta}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi}(i^2 - I_{dc}^2 - 2i\,I_{dc})\,d\theta}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2\,d\theta + \frac{1}{2\pi}\int_0^{2\pi} I_{dc}^2\,d\theta - \int_0^{2\pi} 2i\,I_{dc}\,d\theta} \quad \text{... (13)}$$

First term $$\frac{1}{2\pi}\int_0^{2\pi} i^2\,d\theta = (I_{RMS})^2 \quad \text{... (14)}$$

Second term $$\frac{1}{2\pi}\int_0^{2\pi} I_{dc}\,d\theta = \frac{I_{dc}^2}{2\pi}[\theta]_0^{2\pi}$$

$$= \frac{I_{dc}^2}{2\pi}[2\pi - 0]$$

$$= I_{dc}^2 \quad \text{... (15)}$$

Third term $$\frac{1}{2\pi}\int_0^{2\pi} i^2\,I_{dc}\,d\theta = \frac{2I_{dc}}{2\pi}\int_0^{2\pi} i\,d\theta$$

$$= 2I_{dc}^2 \quad \text{... (16)}$$

Substituting the value of the three terms in equation 13.

$$I'_{RMS} = \sqrt{I_{RMS}^2 + I_{dc}^2 - 2I_{dc}^2}$$

$$I'_{RMS} = \sqrt{I_{RMS}^2 - I_{dc}^2} \quad \text{... (17)}$$

RMS value of a.c. components I'RMS

Now $$\gamma = \frac{\text{RMS value of a.c. components}}{\text{d.c. components}} = \frac{I'_{RMS}}{I_{dc}}$$

$$\gamma = \sqrt{\frac{I_{RMS}^2 - I_{dc}^2}{I_{dc}}}$$

$$= \sqrt{\frac{I_{RMS}^2 - I_{dc}^2}{I_{dc}^2}}$$

$$\boxed{\gamma = \sqrt{\left(\frac{I_{RMS}}{I_{dc}}\right)^2 - 1}} \quad \text{... (18)}$$

Higher value of γ indicates that the output is not pure and contains more a.c. components.

2.7.1 Value of Ripple Factor for Half Wave Rectifier :

For half wave rectifier $I_{dc} = \frac{I_m}{\pi}$ and $I_{RMS} = \frac{I_m}{2}$

$$\gamma = \sqrt{\left(\frac{I_m/2}{I_m/\pi}\right)^2 - 1}$$

$$= \sqrt{\frac{\pi^2}{4} - 1}$$

$$\boxed{\gamma = 1.21} \quad \ldots (19)$$

$\gamma = 1.21$. This indicates that the value of the a.c. components in the output is more than the d.c. component. That is output is not pure d.c.

2.7.2 Value of Ripple Factor for Full Wave Rectifier :

For full wave rectifier $I_{dc} = \frac{2I_m}{\pi}$ and $I_{RMS} = \frac{I_m}{\sqrt{2}}$

$\therefore$ $$\gamma = \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

$$\boxed{\gamma = 0.48} \quad \ldots (20$$

$\gamma = 0.48$ indicates that the better d.c. is available in full wave rectifier than that available from half wav rectifier.

2.8 RIPPLE FREQUENCY :

If we observe the input and the output waveforms of the half wave rectifier, it is seen that there is on cycle of the output for every one cycle of the input a.c. This means that the ripple frequency is equal to tha of the supply frequency. Hence in half wave rectifier the

ripple frequency $\boxed{f_r = f}$ ... (21

In full wave rectifier, there are two pulsations in the output for every one cycle of the input. Hence ripple frequency is twice than that of the supply frequency. Hence

ripple frequency $\boxed{f_r = 2f}$ ... (22

2.9 RECTIFICATION EFFICIENCY :

All the a.c. power fed to the rectifier is not converted into d.c. power. Some power is wasted. Hence efficiency

$$\eta = \frac{\text{Output d.c. power}}{\text{Input d.c. power}} \times 100 \quad \ldots (23$$

$$\boxed{\eta = \frac{P_{dc}}{P_{ac}} \times 100} \quad \ldots (24$$

In this the losses in the transformer are not considered. If these losses are taken into account. then it is called the overall efficiency. Overall efficiency is less than the rectification efficiency.

For half wave rectifier,

$$\text{Output d.c. power} = P_{dc} = I_{dc}^2 R_L$$

$$= \left(\frac{I_m}{\pi}\right)^2 R_L \quad \ldots (25)$$

and $$\text{Input R.C. power} = P_{ac} = I_{RMS}^2 (r_d + R_L)$$

$$= \left(\frac{I_m}{2}\right)^2 (r_d + R_L) \quad \ldots (26)$$

where, r_d = forward resistance of diode

$$\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{(I_m/\pi)^2 R_L}{(I_m/2)^2 (r_d + R_L)}$$

$$= \frac{4}{\pi^2} \frac{R_L}{r_d + R_L} \times 100 \quad \ldots (27)$$

$$\boxed{\eta = \frac{40.6}{\left(\frac{r_d}{R_L} + 1\right)}} \quad \ldots (28)$$

If value of r_d is very small compared to R_L, then $\frac{r_d}{R_L}$ can be neglected and η = 40.6%. That means the maximum efficiency of a half wave rectifier is 40.6%.

For full wave rectifier,

$$\text{Output d.c. power} = P_{dc} = I_{dc}^2 R_L$$

$$= \left(\frac{2I_m}{\pi}\right)^2 R_L \quad \ldots (29)$$

and $$\text{Input R.C. power} = P_{ac} = I_{RMS}^2 (r_d + R_L)$$

$$= \left(\frac{I_m}{\sqrt{2}}\right)^2 (r_d + R_L) \quad \ldots (30)$$

$$\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{(2I_m/\pi)^2 R_L}{(I_m/\sqrt{2})^2 (r_d + R_L)} \times 100$$

$$= \frac{8/\pi^2}{\left(\frac{r_d}{R_L} + 1\right)} \times 100$$

$$\boxed{\eta = \frac{81.2}{\left(\frac{r_d}{R_L} + 1\right)}} \quad \ldots (31)$$

If value of r_d is very small compared to R_L, $\frac{r_d}{R_L}$ can be neglected so η = 81.2%. That means the maximum efficiency of a full wave rectifier is 81.2%.

2.10 PEAK INVERSE VOLTAGE :

In the rectifier circuit, when diode is forward biased, its forward resistance is very less compared to the load resistance. So most of the d.c. voltage is available across the load. But when the diode is reverse biased, its reverse resistance becomes very high in comparison to the load resistance, and all voltage is dropped across the diode. Anode is negative w.r.t. cathode that means the diode gets reverse voltage. The peak value (maximum value) of this reverse voltage is called the peak inverse voltage. Hence peak inverse voltage is the maximum value of the voltage appearing across the diode when it is not conducting.

When the diode is selected for the rectifier, one should know the value of the peak reverse voltage which will appear across the diode depending upon the secondary voltage rating of the transformer and the type of the rectifier circuit. Because the diode selected should bear this peak reverse voltage, since if the diode is not capable to withstand this voltage, breakdown of the junction occurs. Diode selected should have peak inverse voltage (PIV) rating much more than the calculated value.

2.10.1 Value of PIV for Half Wave Rectifier :

Refer to Fig. 2.36(a). It is seen that when diode does not conduct during the negative half cycle, the diode gets reverse voltage E. This is the rms value. So its peak value becomes $\sqrt{2}E$. Hence in half wave rectifier

$$\boxed{PIV = \sqrt{2}E = E_m} \quad \ldots (32)$$

For example if the secondary voltage rating of the transformer is 100 V, then

$$PIV = \sqrt{2}E \times 100 = 141.4 \text{ V}$$

So the *PIV* rating of the diode should be more than 141.4 V so that it can work satisfactorily without giving way in the negative half cycle.

2.10.2 Value of PIV for Full Wave Rectifier with Two Diodes :

Refer to Fig. 2.40(a). Let E volt be available across terminals 'a' and 'c'. During positive half cycle when diode D_1 conducts, diode D_2 does not. Refer to the equivalent circuit shown in Fig. 2.45. Hence diode D_2 gets reverse voltage equal to $E + E = 2E$ during this non conduction period.

Its peak value is $= \sqrt{2}(2E)$

$$= 2E_m \quad \ldots (33)$$

Hence in full wave rectifier with two diodes

$$\boxed{PIV = 2E_m = 2\sqrt{2}E} \quad \ldots (34)$$

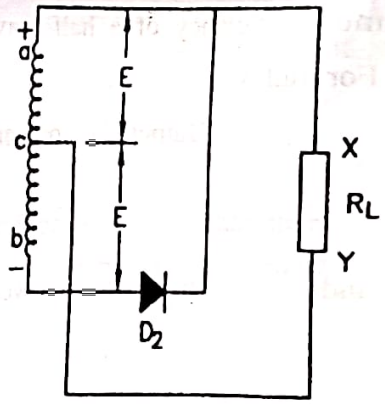


FIG. 2.45 : EQUIVALENT CIRCUIT OF FIG. 2.40(a) WHEN DIODE D_1 IS CONDUCTING

2.10.3 Value of PIV in Full Wave Bridge Rectifier :

To understand the *PIV* in this type of rectifier, the circuit of Fig. 2.41(a) is redrawn here and is shown in Fig. 2.46.

During the positive half cycle terminal 'a' becomes positive w.r.t. 'b', diodes D_1 and D_3 conduct. So these can be replaced by short in the equivalent circuit shown in Fig. 2.47.

It is seen that the two diodes D_2 and D_4 come in parallel to the secondary winding when they do not conduct. Hence the maximum value of this voltage is $\sqrt{2}E$. Therefore in bridge rectifier,

$$\boxed{PIV = \sqrt{2}E = E_m} \quad \ldots (35)$$

as in the half wave rectifier.

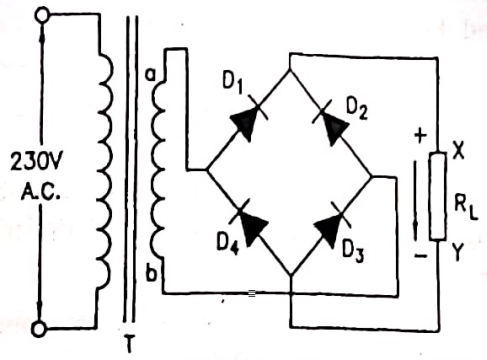


FIG. 2.46 : FULL WAVE BRIDGE RECTIFIER

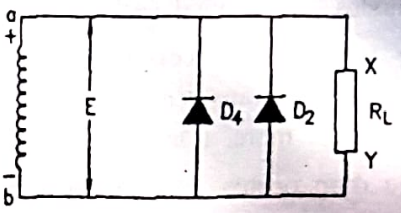


FIG. 2.47 : EQUIVALENT CIRCUIT OF FIG. 2.46 DURING THE POSITIVE HALF CYCLE

2.11 FORM FACTOR :

It is the ratio of the RMS value and the average value.

$$\text{Form factor} = \frac{RMS \text{ value}}{\text{average value}} \quad \ldots (36)$$

For half wave rectifier,

$$I_{RMS} = \frac{I_m}{2} \text{ and } I_{dc} = \frac{I_m}{\pi}$$

$$\therefore \quad \text{Form factor} = \frac{I_m/2}{I_m/\pi} = \frac{\pi}{2} = \boxed{1.57} \quad \ldots (37)$$

And for full wave rectifier,

$$I_{RMS} = \frac{I_m}{\sqrt{2}} \text{ and } I_{dc} = \frac{2I_m}{\pi}$$

$$\therefore \quad \text{Form factor} = \frac{I_m/\sqrt{2}}{2I_m/\pi} = \frac{\pi}{2\sqrt{2}} = \boxed{1.11} \quad \ldots (38)$$

2.12 COMPARISON OF RECTIFIER CIRCUITS :

2.12.1 Comparison of Half Wave Rectifier and Full Wave Rectifier with Two Diodes :

Half Wave Rectifier	Full Wave Rectifier with Two Diodes
1. Only one diode is required (Adv.)	1. Two diodes are necessary. (Disad.)
2. No centre tapping is necessary in the secondary winding of the transformer. (Adv.)	2. Transformer with centre tapping in the secondary winding is required. (Disad)
3. D.C. output voltage $E_{dc} = \frac{E_m}{\pi}$ which is half of that in full wave rectifier. (Disad)	3. D.C. output voltage $E_{dc} = \frac{2E_m}{\pi}$ which is double than half wave rectifier. (Adv.)
4. $I_{RMS} = \frac{I_m}{2}$	4. $I_{RMS} = \frac{I_m}{\sqrt{2}}$
5. PIV = E_m (Adv.)	5. PIV = $2E_m$ (Disad.)
6. Efficiency = 40.6% Less (Disad)	6. Efficiency = 81.2% more (Adv.)
7. Ripple frequency = f	7. Ripple frequency = $2f$
8. Ripple factor = 1.21 (Disad)	8. Ripple factor = 0.48 (Adv.)
9. Costly filters are necessary	9. Comparatively less costly filters are necessary. (Adv.)

2.12.2 Comparison of Full wave Rectifier with Two Diodes and Full Wave Bridge Rectifier :

Full Wave Rectifier with Two Diodes	Full Wave Bridge Rectifier
1. *Two diodes* are required (Adv.)	1. Four diodes are required (Disad).
2. Centre tapping is necessary in the secondary winding of the transformer. So the cost of the transformer is more. (Disad)	2. No centre tapping is necessary in the secondary winding of the transformer. So cost of the transformer is less. (Adv.).
3. Voltage drop occurs only in one diode so drop in output voltage is less (Adv.).	3. Two diodes are connected in series at any time so the voltage drop is twice (Disad).
4. Peak inverse voltage is $2E_m$ (Disad).	4. Peak inverse voltage is E_m (Adv.).

2.12.3 Comparison of Three Types of Rectifiers :

Sr. No.	Item	Half wave rectifier	Full wave rectifier with two diodes	Full wave bridge rectifier
1.	No. of diodes	1	2	4
2.	Need of centre tapping in secondary winding of the transformer	No	Yes	No
3.	Average d.c. voltage E_{dc}	E_m/π	$2E_m/\pi$	$2E_m/\pi$
4.	Average d.c. current I_{dc}	I_m/π	$2I_m/\pi$	$2I_m/\pi$
5.	RMS value of current I_{RMS}	$I_m/2$	$2I_m/\sqrt{2}$	$I_m/\sqrt{2}$
6.	Ripple factor γ	1.21	0.48	0.48
7.	Ripple frequency	f	$2f$	$2f$
8.	Maximum rectification efficiency η	40.6%	81.2%	81.2%
9.	Average diode current	I_{dc}	$I_{dc}/2$	$I_{dc}/2$
10.	Form factor	1.57	1.11	1.11
11.	Filtering	difficult	easy	easy
12.	Voltage drop in diode	less	less	more
13.	Application	less	used for low voltage	used for high voltage

2.13 FILTERS IN RECTIFIER :

Necessity of filter :

Output current from the rectifier is pulsating. It is the mixture of the d.c. component and the a.c. components. Filter circuit is used to remove these a.c. components and to get smooth d.c. Hence by using filter circuit, a.c. components are removed and smooth d.c. is obtained. Capacitor, inductor and their combinations are used in the filter circuits.

Types of filter circuits :

The following types of the filter circuits are available :

1. Shunt capacitor filter
2. Series inductor filter

3. Choke input *LC* filter
4. Capacitor input *LC* filter
5. π filter

2.13.1 Shunt Capacitor Filter :

As shown in Fig. 2.48 a capacitor of higher value is connected across the output terminals of the rectifier.

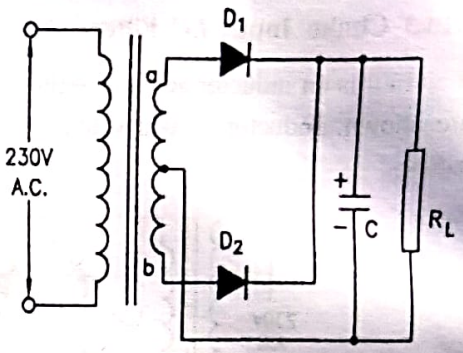


FIG. 2.48 : SHUNT CAPACITOR FILTER

Action of the capacitor can be understood by two ways.

Capacitive reactance of a capacitor is given by $X_C = \frac{1}{2\pi f C}$. So it offers less reactance to the a.c. components (due to their frequency). So these components are bypassed to ground and will not flow through the load. Frequency of d.c. is zero, so the capacitor offers infinite reactance to the d.c. components. So d.c. component can not pass through the capacitor and has to pass through the load.

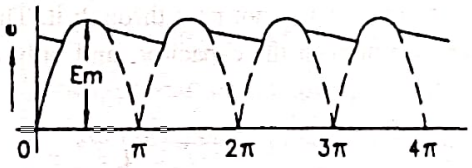


FIG. 2.49 : OUTPUT VOLTAGE WAVEFONN OF SHUNT CAPACITOR FILTER

The action of the capacitor can be understood in the other way also. Capacitor is charged to the peak value E_m.

When the d.c. voltage decreases, the capacitor discharges through the load and tries to maintain the load current. However due to discharging load voltage decreases slightly. This depends on the value of the capacitor. If the value of the capacitor is not enough the decrease in load voltage is more and if the value of the capacitor is large enough the decrease in the load voltage is less. In the next half cycle the capacitor again charges to E_m. When capacitor filter is used, the output d.c. voltage is approximately E_m. Waveforms of output voltage are shown in Fig. 2.49.

2.13.2 Series Inductor Filter :

Current flowing through the inductor can not change instantaneously. If the current tries to decrease, emf is induced in the inductor which opposes the decrease in current. Series inductor filter works on this principle.

In this an inductor L (or choke) is connected in series as shown in Fig. 2.50. Inductive reactance of the inductor is

$$X_L = 2\pi f L$$

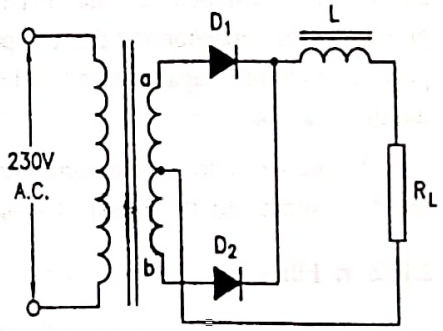


FIG. 2.50 : SERIES INDUCTOR FILTER

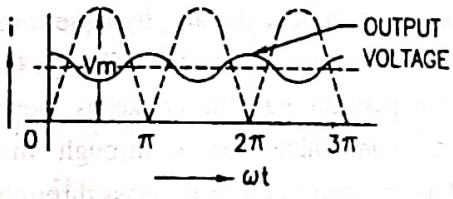


FIG. 2.51 : WAVEFORMS OF SERIES INDUCTOR FILTER

Frequency of d.c. is zero, so it offers zero reactance to the d.c. components. But it offers more reactance to the a.c. components. So d.c. component passes through the inductor and load. The a.c. components are blocked by the inductor and can not pass through the load. Some voltage is dropped across the inductor so the output voltage is decreased. Some ripple is present in the output. The waveforms are shown in Fig. 2.51.

This filter works on the current principle. So it is not used with the half wave rectifier as there is no current during the negative half cycles.

2.13.3 Choke Input *LC* Filter :

In this an inductor and a capacitor are used. In Fig. 2.52(a) the circuit is shown and in figure (b), the waveforms are shown. Inductor is connected in series and capacitor is connected in parallel.

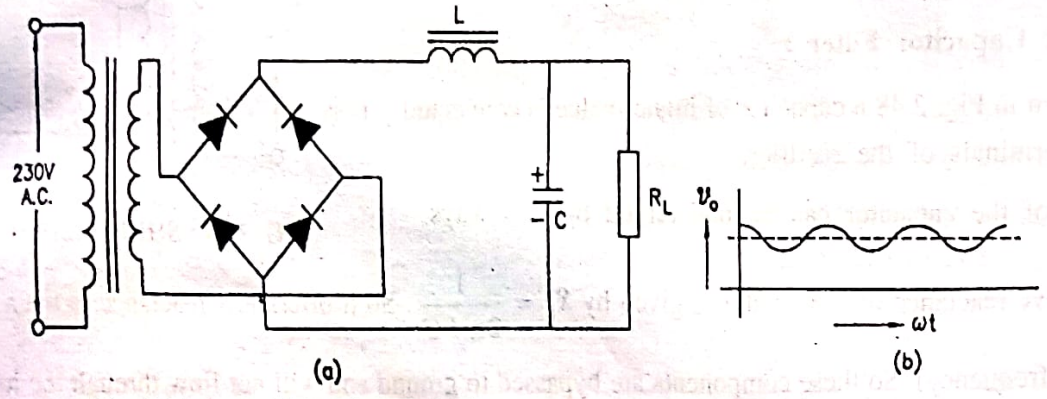


FIG. 2.52 : CHOKE INPUT L_C FILTER (a) CIRCUIT (b) WAVEFORMS

First the choke is connected and then the capacitor. So this type of filter is known as the choke input *LC* filter.

Inductor allows the d.c. component to pass through it where-as a.c. components can not pass through it. Those low frequency a.c. components which pass through the inductor are bypassed through the capacitor, and only d.c. passes through the load.

2.13.4 Capacitor Input *LC* Filter :

In this type of filter, first the capacitor is connected and then the series inductor (choke) is connected as shown in Fig. 2.53.

In this a.c. components are opposed by the choke but d.c. component passes easily through the choke. A.C. components are bypassed to the ground by the capacitor and d.c. component cannot pass through the capacitor and only d.c. passes through the load.

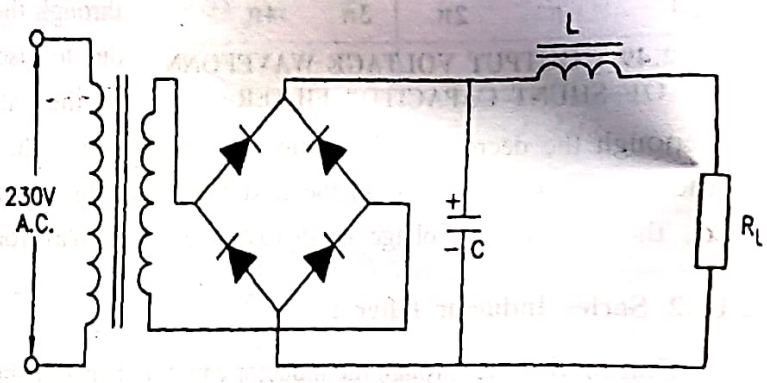


FIG. 2.53 : CAPACITOR INPUT LC FILTER

As the capacitor is connected immediately after the rectifier, the capacitor is charged to the peak value of the d.c. output. So there is less drop of voltage. Waveforms are similar to that shown in Fig. 2.53(b).

2.13.5 π Filter :

In this, first shunt capacitor C_1 is connected, then the series choke is connected. Then second shunt capacitor C_2 is connected. Circuit is shown in Fig. 2.54.

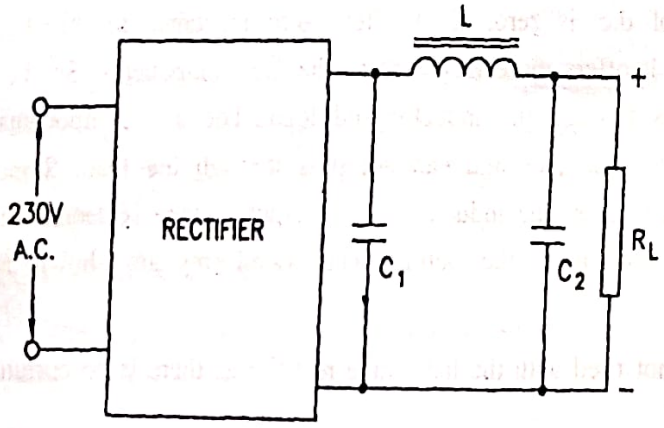


FIG. 2.54 : π TYPE FILTER

Most of the a.c. components are bypassed to ground by the first shunt capacitor C_1. A.C. components can not pass through the choke as these are impeded. D.C. component passes through the inductor and reach to the load as it can not pass through the shunt capacitor C_2. Those low frequency a.c. components which could pass through the choke are bypassed to ground by another shunt capacitor C_1 and it does not allow these a.c. components to pass through the load.

There is some voltage drop in the inductor.

This type of filter circuit is known as the π type filter because the configuration of the filter resembles to the greek letter π.

Sometimes in this type of filter a resister is used in place of choke. Due to this there is saving in the cost but it increases the voltage drop and there is power loss in the resistor.

2.13.6 Comparison Between the Choke Input and the Capacitor Input *LC* Filter :

In both the types of filters, one choke and one capacitor are used.

In choke input filter first the series choke is connected and then the capacitor is connected in parallel. Whereas in the capacitor input type *LC* filter, first the shunt capacitor is connected and then the choke is connected in series.

In the choke input filter the current of the rectifier passes through the choke and so there is voltage drop across the choke. So output voltage is reduced. However the voltage regulation is good for more load current.

In capacitor input filter, the capacitor charges to the peak value of the output voltage. Also the a.c. components are bypassed to ground so there is less voltage drop in the choke. Output voltage is more than that in the choke input *LC* filter.

2.14 APPLICATIONS OF RECTIFIERS :

Some applications of low and medium power rectifiers are mentioned below :

1. Radio receiver circuits work on d.c. supply.
2. Different stages of TV receiver require d.c. supply.
3. Electronic equipments like audio generator, amplifier, CRO etc. need d.c. supply for their operation.
4. It is used in battery charging.
5. It is used in electroplating on small scale.

2.15 DC AC POWER SUPPLY :

A power supply is a device that converts the electrical energy coming from a power source into the voltage and current values powering a load. In other words, the power supply ensures that electrical power is supplied in a controlled manner. A power supply can control and accurately adjust the voltage to several loads sometimes simultaneously without allowing changes in input voltage or other connected devices to affect the output.

2.15.1 Difference Between AC and DC Power Supply :

Alternating current (AC) power supply refers to electric charges that change direction periodically. The voltage in AC circuits also regularly reverses due to current direction changes. AC power supply is typically used to power office buildings, homes, and more.

On the other hand, direct current (DC) power supplies provide a constant voltage or current. The current from a DC power supply only flows in one direction. This type of power supply is typically used for small electronics and devices that use a battery.

2.15.2 Direct Current and Alternating Current :

Direct current : In direct current, the direction of current does not change. The value of current can be increased or decreased.

In Fig. 2.55, graph of current versus time is shown. There is no change in magnitude of current in curve 'a', but in curve 'b', the magnitude of current decreases as the timer passes. But in both, the direction of current does not change. So, both are the examples of direct current.

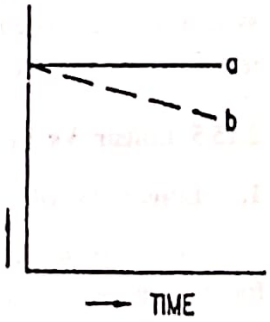


FIG. 2.55

In Fig. 2.56 and 2.57 both are the examples of direct current. There is no change in the direction of current, but the magnitude of current goes on changing. Both are called the pulsating a.c. The output from the half wave rectifier has the waveform as shown in Fig. 2.56 while output from the full wave rectifier has the waveform is as shown in Fig. 2.57.

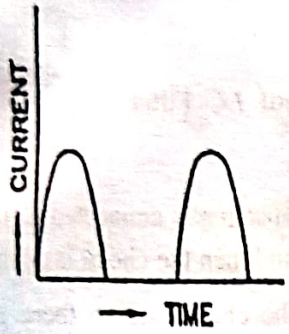


FIG. 2.56

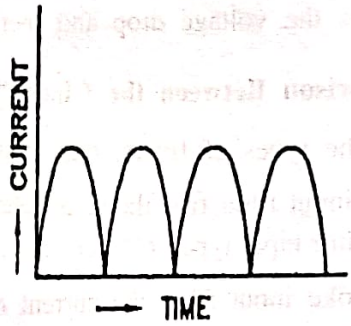


FIG. 2.57

Alternating current : In alternating current the magnitude and the direction of current continuously goes on changing. The graph of alternating charging as per sinewave is shown in Fig. 2.58.

It is seen from the figure that the value of current increases from zero in one direction and becomes maximum, then decreases slowly to zero. Then it increases in negative direction, becomes maximum, then becomes zero. The cycle then repeats.

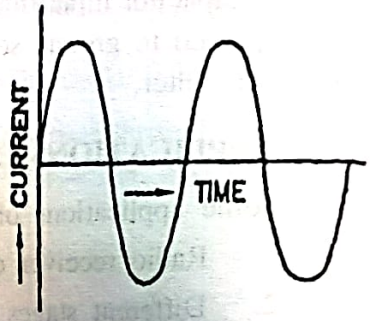


FIG. 2.58

2.15.3 Sources of Electricity :

There are two sources of electricity :

(1) Source producing direct current and (2) Source producing alternating current.

There are two types of sources producing direct current : (1) battery and (2) d.c. generator.

Battery : Battery converts chemical energy in to electrical energy. Battery is the source of direct current.

There are two terminals namely positive and negative. When load is connected to the battery, the current flows from positive terminal to the load and back to the negative terminal.

D.C. generator : This is the rotating electrical machine. The main parts are armature, field and commutator. Mechanical energy is converted in to electrical when the armature is rotated by the prime mover. D.C. generator generates electricity in the form of direct current.

Alternator : Alternator is also a rotating electrical machine. It is also called a.c. generator. It generates electricity in the form of a.c. In this also, the main parts are the armature and the field. The armature or field is rotated by the prime mover (steam turbine or water turbine) so the mechanical energy is converted in to electrical energy.

2.15.4 A.C. and D.C. Systems :

Initially only d.c. system was prevailing. Then due to the invention of transformer and development, a.c. system was developed. Today a.c. system is widely used. D.C. is highly essential in some applications. For this, rectifier is used to convert a.c. in to d.c.

2.15.5 Linear Vs Switching AC/DC Power Supply :

1. Linear AC/DC Power Supply :

Linear AC/DC power supplies use a simple design. The AC input voltage can be reduced to a suitable value for the intended application by using a transformer. With the voltage reduced, it's then rectified and turned into DC voltage. From there, it's filtered to further improve the waveform quality.

One hindrance of the linear AC/DC power supply is the transformer's size. Since the input voltage is transformed at the input, the transformer would need to be large and heavy. Large inductance-values are necessary to transfer high amounts of power from the primary to the secondary coil at low frequencies. This makes it nearly impossible to make miniature power supplies.

To maintain a consistent voltage at the output, a linear AC/DC power supply uses linear regulators. Any extra energy will be dissipated in the form of heat by the linear regulators. At high power, the regulator would need to dissipate a lot of heat to maintain a constant output voltage. It would need an extremely large heat sink.

2. Switching AC/DC Power Supply :

Modern semiconductor technology can handle power supply switching on and off quickly and efficiently even when large voltages and currents are present. A switching AC/DC power supply allows for the elimination of dissipating excess power since it works more efficiently.

AC/DC power supplies use converters called switched mode power supplies. These use a complex method for converting AC to DC power. In switching AC power supplies, there's no need to reduce input voltage it's rectified and filtered in the input. The DC voltage travels through a chopper, where it converts the voltage into a high frequency pulse train. After that, the wave will go through another rectifier and filter, which converts it back to DC and eliminates the remaining AC component.

2.16 REGULATED POWER SUPPLY :

We studied voltage regulator circuit. This circuit is used to give constant voltage to the circuit and is used in regulated power supply. Here we shall obtain information about regulated power supply.

2.16.1 Block Diagram of Regulated Power Supply :

Block diagram of regulated power supply is given in Fig. 2.59.

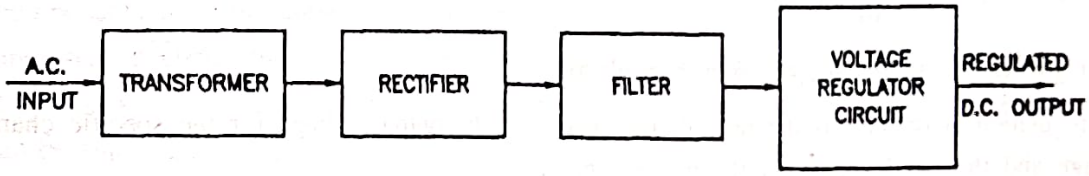


FIG. 2.59 : BLOCK DIAGRAM OF REGULATED POWER SUPPLY

Functions of different blocks are given below.

(1) **Transformer :** Stepup or step down transformer is used to get the required a.c. voltage.

(2) **Rectifier :** Alternating voltage is converted in to d.c. voltage with the help of rectifier. Full wave type rectifier is used.

(3) **Filter circuit :** Output voltage from rectifier is pulsating d.c. This pulsating d.c. voltage is made smooth d.c. with the help of filter circuit.

(4) **Voltage regulator circuit :** The output voltage changes due to the change in the input a.c. voltage and load current. Voltage regulator circuit keeps the output voltage constant. Over load and short circuit protection are provided in this circuit.

2.16.2 Load Regulation :

From the load regulation we can know what will be the change in the output voltage when the load current changes from no load to the full load value. When the load current is zero, there is no voltage drop and the output voltage is the maximum and when the power supply gives full load current there is reduction in the output voltage.

If V_{NL} = Output voltage at zero load current (No load output voltage)

V_{FL} = Output voltage at full load (Full load output voltage), then

$$\text{Load regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}}$$

$$\text{And \% load regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100$$

This can be defined as follows.

Percentage load regulation is defined as the change in the output voltage from no load to full load expressed as the percentage of the no load voltage.

For example, if the no load output voltage of a power supply is 24 V and if the output voltage falls to 23 Vat full load, then

$$\text{\% Load regulation} = \frac{24 - 23}{24} \; 100 = 4.17 \; \%$$

2.16.3 Line Regulation or Source Regulation :

Let us suppose that in a power supply 10 V d.c. output is available when the input a.c. voltage is 230 V. If there is change in the input a.c. voltage, there is change in the d.c. output voltage. Suppose the output voltage becomes 9.8 V when the input voltage becomes 210 V and the output voltage becomes 10.2 V when the input voltage becomes 250 V. That is the output voltage changes from 9.8 V to 10.2 V when the input voltage changes from 210 V to 250 V. So there is change of 10.2 – 9.8 = 0.4 V and the rated output voltage is 10 V. So

$$\text{Line regulation} = \frac{10.2 - 9.8}{10} \times 100 = 4 \; \%$$

Line regulation is also called the source regulation.

Line regulation is defined as the ratio of the change in the output voltage for the specific change in the input voltage and the rated voltage of the power supply.

$$\text{\% Line regulation} = \frac{\text{Change in the output voltage for the specific change in the input voltage}}{\text{Rated voltage}}$$

2.17 SPECIFICATIONS OF REGULATED POWER SUPPLY :

Important specifications of regulated power supply are as follows.

(1) **Output voltage :** Output voltage means the d.c. regulated output voltage available from power supply. This voltage is constant or its value may be changed as required. For example, 15 V or 0 to 30 V.

(2) **Output current :** This indicates the maximum value of the current which can be taken from the power supply. For example, 500 mA, 1 A, 2 A, 5 A etc.

(3) **Load regulation :** It shows the change in the output voltage when the load current becomes maximum, from zero value. For example, 0.1%, 0.05% etc.

(4) **Line regulation or source regulation :** Change in the output voltage for the specific change in the input a.c. voltage is called the line regulation. For example, 0.2%, 0.08% etc.

(5) **Percentage ripple :** Small magnitude of ripple still remains in the output d.c. voltage even after filtering. This ripple is shown in percentage. For example, 0.02%, 0.005% etc.

(6) **Protection :** It is necessary to indicate what type of protection is given in the power supply. For example, over load protection, short circuit protection etc.

In addition to this, the type of enclosure, meters like ammeter, volt mete etc. are provided with the power supply and if provided the type say analog type or digital type etc. are also required to be mentioned.

2.18 COMPLETE CIRCUIT DIAGRAM OF REGULATED POWER SUPPLY USING SERIES REGULATOR :

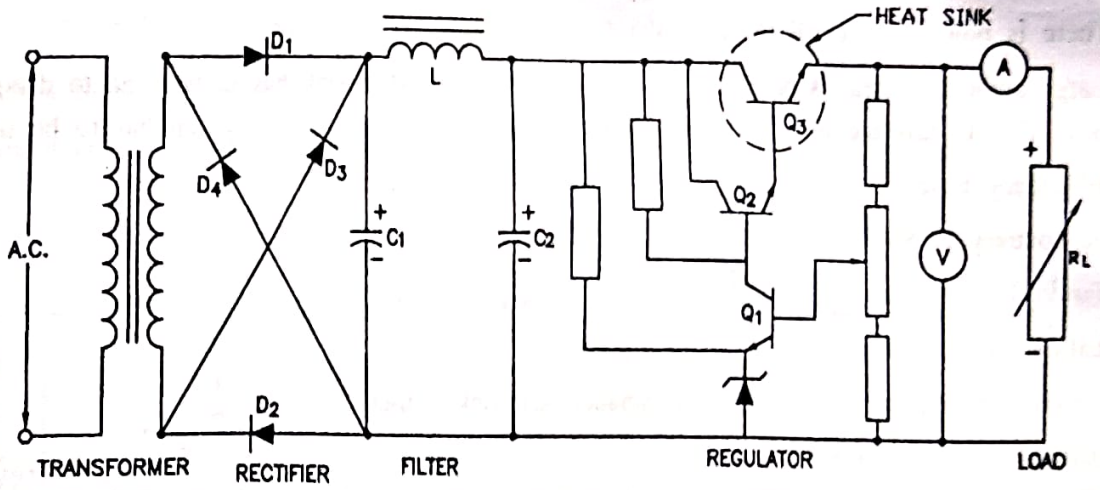


FIG. 2.60 : COMPLETER CIRCUIT DIAGRAM OF REGULATED POWER SUPPLY USING SERIES REGULATOR

Complete circuit diagram of regulated power supply is shown in Fig. 2.60. Required a.c. voltage is obtained with the help of transformer. Bridge type rectifier converts a.c. voltage in to d.c. voltage. Filter circuit removes the a.c. components from the output. Three transistors are used in series type voltage regulator. Transistors Q_2 and Q_3 are used as Darlington pair to increase the current rating. Transistor Q_1 is the comparator. Its base is given the feedback voltage and its emitter is given the reference voltage with the help of zener diode. Output voltage can be adjusted by the potentiometer. Output voltage can be measured by the voltmeter and the load current can be measured by the ammeter.

Overload and short circuit protection are not shown in the above circuit but these are invariably provided in the commercial power supplies.

2.19 COMPLETE CIRCUIT OF A REGULATED POWER SUPPLY USING SHUNT REGULATOR :

Fig. 2.61 shows the complete circuit diagram of regulated power supply using the shunt regulator.

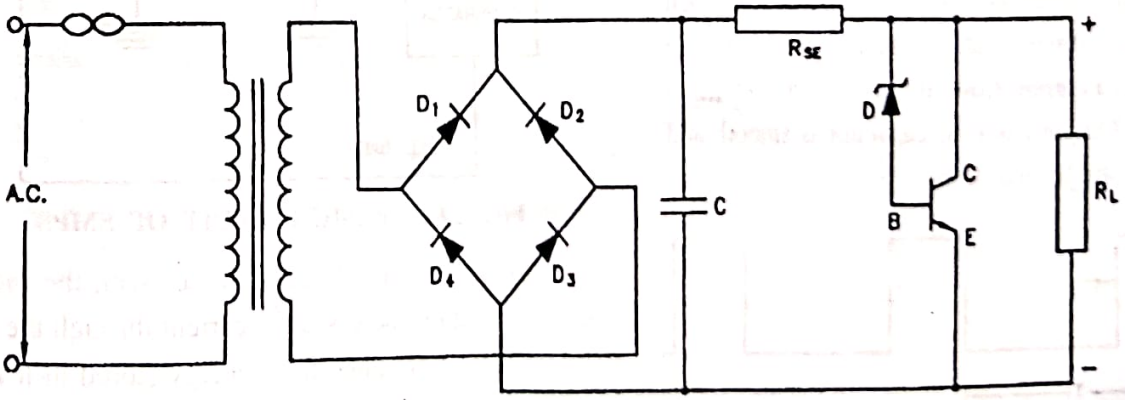


FIG. 2.61 : COMPLETER CIRCUIT DIAGRAM OF A REGULATED POWER SUPPLY USING SHUNT REGULATOR

Supply voltage is stepped down to the desired value with the help of transformer. Bridge rectifier converts a.c. into pulsating d.c. A.c. components in the output are removed by capacitor C and smooth d.c. is obtained. Shunt regulator comprising of zener diode D and shunt regulator stabilizes the output voltage.

2.20 SWITCHED MODE POWER SUPPLY – SMPS :

Power supplies using series pass transistors were popular up to 1980. These are called linearly regulated power supply (LRPS). Nowadays switched mode power supplies have become popular. The following are the disadvantages of power suppliers using series pass transistor.

(1) There is power loss equal to $(V_i - V_o)\ I_L$ in the series pass transistor.

(2) Large amount of heat is produced due to this power loss. Heat sink has to be used to dissipate this heat. Due to this the power supply becomes bulky. Sometimes the cooling fan has to be used.

(3) Efficiency is less.

On the contrary in SMPS

(1) Power loss is less.

(2) Efficiency is more.

(3) Power supply becomes less bulky as smaller heat sink is used.

Comparison is given in table

LRPS	SMPS
1. There is power loss equal to $(V_i - V_o)\ I_L$	1. Power loss is less.
2. More heat is produced due to power loss.	2. Less heat is produced.
3. Larger heat sink has to be used.	3. Smaller heat sink is needed.
4. Due to this the power supply becomes bulky.	4. Weight of power supply is loss.
5. Efficiency is less.	5. Efficiency is more.
6. Circuit is simple	6. Circuit is complicated.
7. Cost is µm.	7. Cost is more.

2.20.1 Basic Circuit of SMPS :

Basic circuit of SMPS is shown in Fig. 2.62.

Transistor Q works as a switch and it is driven by the pulse generator. Pulse generator produces pulse train. Transistor goes in to saturation when the pulse comes and produces pulse in the collectors so current flows through the inductor and capacitor. Output from the capacitor is smooth and the load can be fed from this.

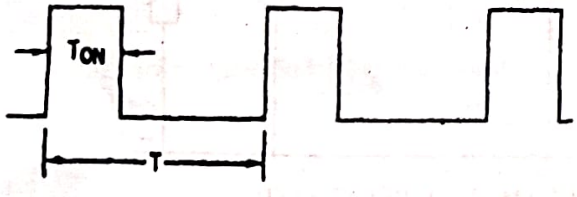


FIG. 2.63 : WAVEFORMS OF PULSE GENERATOR

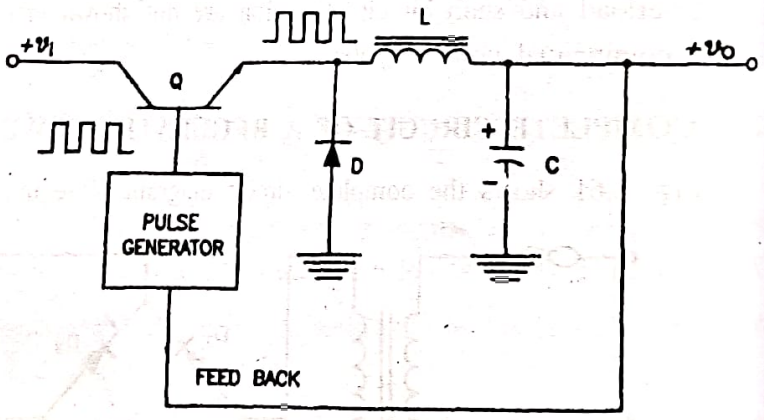


FIG. 2.62 : BASIC CIRCUIT OF SMPS

When the output voltage decreases with the increase in load, pulse width is increased. So current through the inductor flows for some more time so the energy stored in it increases and the voltage of the capacitor Increases. So the load voltage increases.

Similarly, when the output voltage increases, the pulse width is decreased so the output voltage decreases. Thus the output voltage remains constant.

Duty $D = \frac{T_{ON}}{T}$

If the input voltage is V_i and the output voltage is V_o,

$$V_o = V_i \, D$$

For example, if the input voltage is 15 V and the duty is 0.6, then

Output voltage is $V_o = 15 \times 0.6 = 9$ V and if duty is 0.4 then $V_o = 15 \times 0.4 = 6$ V.

2.20.2 Block Schematic or SMPS :

Block diagram of one type of SMPS is shown in Fig. 2.64.

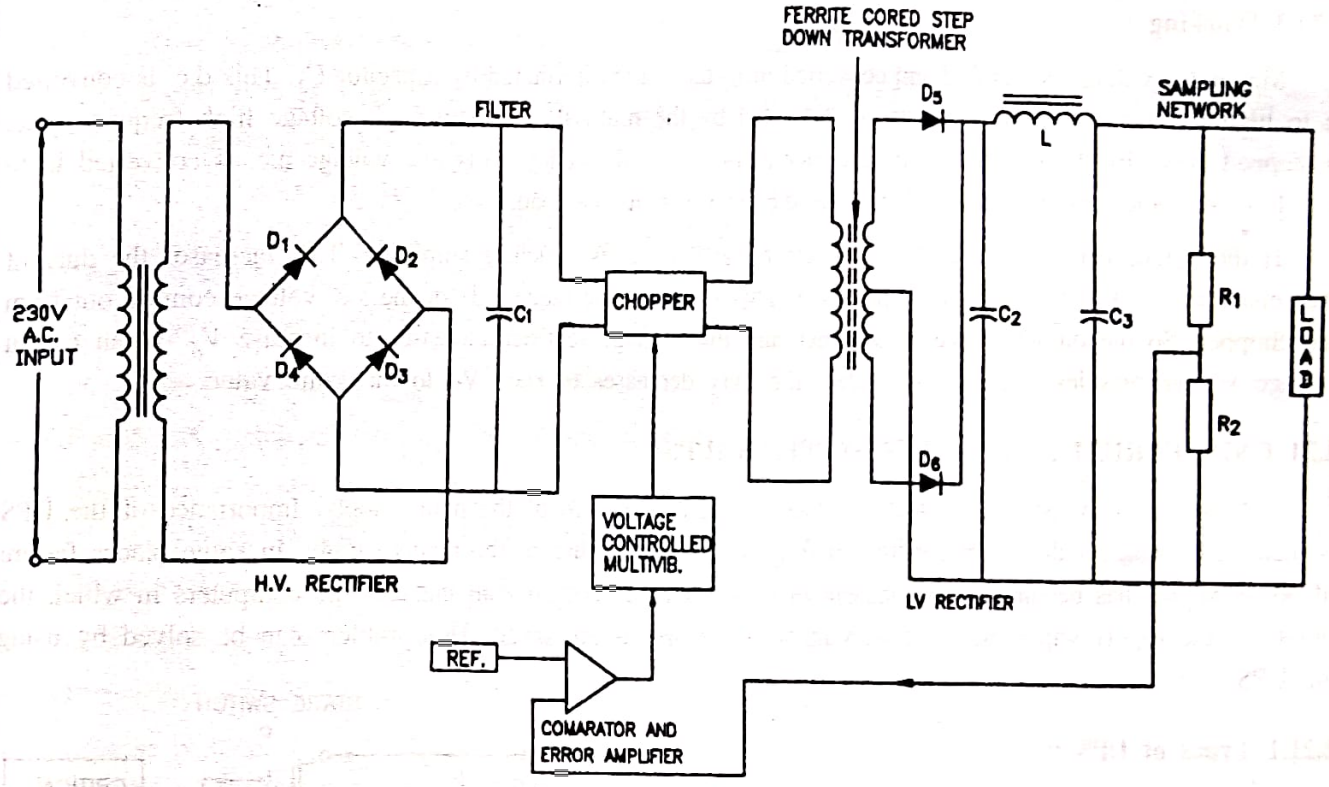


FIG. 2.64 : BLOCK DIAGRAM OF SMPS

Function of various blocks are as below :

1. **H.V. rectifier :** It is a bridge rectifier comprising of high voltage diodes D_1 through D_4. Mains a.c. voltage is converted in to d.c.
2. **Filter :** Shunt capacitance filter C_1 smoothes out the pulsating d.c.
3. **Chopper :** Chopper converts high voltage d.c. into high voltage high frequency a.c. The magnitude of the a.c. voltage is increased or decreased depending upon the output voltage.
4. **Step down transformer :** It is a ferrite cored step down transformer. It steps down the high voltage high frequency a.c. into low voltage a.c. at the same frequency. Since the frequency is high (20 kHz to 50 kHz), the size of the transformer is very much reduced compared to mains step down transformer which is used in the conventional power supplies.
5. **LV rectifier :** It is a full wave type rectifier comprising of diodes D_5 and D_6. It converts low voltage a.c. into d.c.

6. **Filter :** It is a π type filter comprising of capacitors C_2, C_3 and choke L. It smoothes out d.c. output from the LV rectifier.
7. **Sampling network :** It comprises of resistors R_1 and R_2 connected in series across the output voltage. The voltage available across R_2 is proportional to the output voltage V_o. This voltage is applied to the comparator.
8. **Reference :** It generates the steady reference voltage. Normally it is obtained by using zener diode.
9. **Comparator and error amplifier :** It receives two signals. One is the reference signal and the other is proportional to the output voltage. The difference of these two signals is the error signal. This error signal is amplified by the amplifier. This amplified signal is given to the voltage controlled multivibrator.
10. **Voltage controlled multivibrator :** It is a square wave generator. Its mark to space ratio or duty is controlled by the error signal. Output of the multivibrator is applied to the chopper.

2.20.3 Working :

Mains a.c. voltage is rectified and converted in to d.c. which is filtered by capacitor C_1. This d.c. is converted in to high frequency a.c. by the chopper controlled by the multivibrator. This high voltage high frequency a.c. is stepped down by the ferrite core transformer at the same frequency. This low voltage a.c. is converted in to d.c. It is smoothed out by the π filter. This d.c. is given to the load.

If the output voltage V_o decreases, the error signal changes which is amplified. This increases the duty of the square wave i.e. the mark time is increased. This increases the magnitude of the a.c. voltage coming out from the chopper. So the output of the transformer and low voltage rectifier increases to increase V_o. When output voltage V_o becomes less than the set value, the duty decreases to keep Vo to the same value.

2.21 UNINTERRUPTABLE POWER SUPPLIES (UPS) :

UPS are used to feed the critical a.c. loads in case of failure of the main supply. Importance of the UPS is increased nowadays due to the, increased frequency of the failure of the mains supply. In some places failure of power supply has become routine instead of exception. The critical load includes the computers in which the failure of the supply wipes off the entries made if the data is not saved. This problem can be solved by using the UPS.

2.21.1 Types of UPS :

In UPS main frequency inverters driven by the storage batteries are used. The batteries are charged during the healthy condition. There are two arrangements of the UPS :

1. Critical load normally connected to the a.c.mains and
2. Critical load connected to the inverter.

In first arrangement as shown in Fig. 2.65 the critical load is connected to the a.c.mains. Storage batteries are charged during the healthy condition. When power fails, the inverters are switched on automatically arid the load is supplied by the inverter.

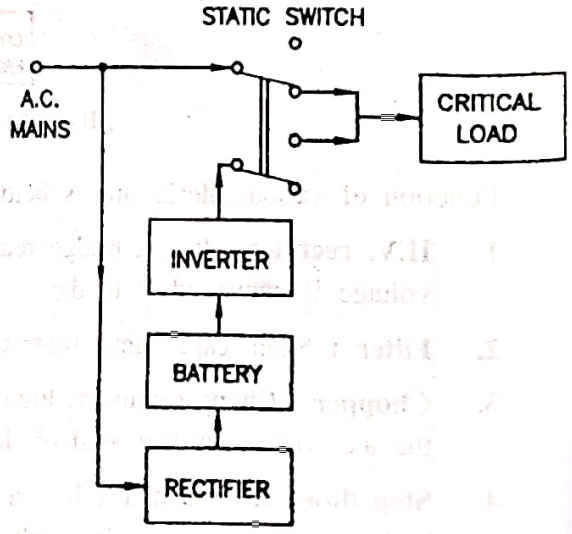


FIG. 2.65 : ONE ARRANGEMENT OF UPS

Features of this arrangement are :

1. Load circuit is broken momentarily for 4 to 5 ms when static switches are used and for 30 to 50 ms when electromechanical relays are used.
2. Inverter runs only during the time when supply failure occurs.

3. The load is not saved from the switching transients of the a.c.mains as it is connected directly to the a.c.mains.

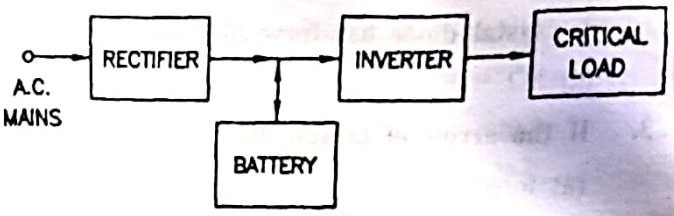


FIG. 2.66 : ANOTHER ARRANGEMENT OF UPS

In the second arrangement as shown in Fig. 2.66, the critical load is connected permanently to the inverter. The battery is charged from the d.c. obtained from the rectifiers. The d.c. of the rectifier also drives the inverter. The inverter supplies the critical load. When the mains a.c. supply fails, the charging process of the battery stops but the inverter continues to run on the battery. Hence there is continuity of the supply both during the healthy and the faulty conditions.

Features of this arrangement are :

1. Critical load is supplied from the inverter.
2. Inverter is continuously on.
3. There is no interruption of the supply to the critical load.
4. The load is protected from the transients of the main supply as the load is not connected directly to the mains.

SUMMARY

1. Diode is bipolar device consisted by P-type and N-type semiconductor.
2. In N-type semiconductor electrons are majority charge carriers and hole are minority charge carriers.
3. In P-type semiconductor holes are majority charge carriers and electrons are minority charge carriers.
4. Diode has a two terminals anode and cathode. Diode is a unidirectional current device.
5. When anode is most positive than cathode it flows current that's called "Forward Bias" whereas anode is most negative than cathode that's called reverse bias. In reverse bias diode cannot flow current in circuit.
6. Rectifier is diode application which converts a.c. supply into d.c. supply.
7. Half wave does rectification of only half cycle of a.c. input supply.
8. Full wave rectifier circuit rectifies both the half cycle of a.c. input supply.
9. Clipper is diode based circuit which removes unwanted wave shape of input signal. There are four types of clipper like positive, negative, biased and combined clipper.
10. Filter circuit uses C and L, for removing a.c. components from rectified d.c. of rectifier circuits.
11. Performance parameters of rectifiers are V_{dc}, V_{rms}, ripple frequency, ripple factor, PIV, rectification efficiency, transformer utilization factors, power factors are used for analysis of rectifier in d.c. power supply.
12. Linear regulated d.c. supply and SMPS are used as regulated power supply.
13. UPS is important AC/DC power supply to run critical electrical load.
14. SMPS has more efficiency than linear regulated power supply.
15. ONLINE and OFFLINE UPS configuration are used as per necessity of supplying the nature of load.

MULTIPLE CHOICE QUESTIONS (MCQs)

1. A crystal diode has _____.

(a) one pn junction
(b) two pn junctions
(c) three pn junctions
(d) none of the above

2. A crystal diode has forward resistance of the order of _____.
(a) kΩ (b) Ω (c) MΩ (d) none of the above

3. If the arrow of crystal diode symbol is positive w.r.t. bar, then diode is _____ biased.
(a) forward (b) reverse
(c) either forward or reverse (d) none of the above

4. The reverse current in a diode is of the order of _____.
(a) kA (b) mA (c) μA (d) A

5. The forward voltage drop across a silicon diode is about _____.
(a) 2.5 V (b) 3 V (c) 10 V (d) 0.7 V

6. A crystal diode is used as _____.
(a) an amplifier (b) a rectifier (c) an oscillator (d) a voltage regulator

7. The d.c. resistance of a crystal diode is _____ its a.c. resistance.
(a) the same as (b) more than (c) less than (d) none of the above

8. An ideal crystal diode is one which behaves as a perfect _____ when forward biased.
(a) conductor (b) insulator (c) resistance material (d) none of the above

9. The ratio of reverse resistance and forward resistance of a germanium crystal diode is about _____.
(a) 1 : 1 (b) 100 : 1 (c) 1000 : 1 (d) 40000 : 1

10. The leakage current in a crystal diode is due to _____.
(a) minority carriers (b) majority carriers (c) junction capacitance (d) none of the above

11. If the temperature of a crystal diode increases, then leakage current _____.
(a) remains the same (b) decreases (c) increases (d) becomes zero

12. The PIV rating of a crystal diode is _____ that of equivalent vacuum diode.
(a) the same as (b) lower than (c) more than (d) none of the above

13. If the doping level of a crystal diode is increased, the breakdown voltage _____.
(a) remains the same (b) is increased (c) is decreased (d) none of the above

14. The knee voltage of a crystal diode is approximately equal to _____.
(a) applied voltage (b) breakdown voltage (c) forward voltage (d) barrier potential

15. When the graph between current through and voltage across a device is a straight line, the device is referred to as _____.
(a) linear (b) active (c) nonlinear (d) passive

16. When the crystal diode current is large, the bias is _____.
(a) forward (b) inverse (c) poor (d) reverse

17. A crystal diode is a _____ device.
(a) non-linear (b) bilateral (c) linear (d) none of the above

18. A crystal diode utilises _____ characteristic for rectification.
(a) reverse (b) forward (c) forward or reverse (d) none of the above

19. When a crystal diode is used as a rectifier, the most important consideration is _____.
(a) forward characteristic (b) doping level
(c) reverse characteristic (d) PIV rating

20. If the doping level in a crystal diode is increased, the width of depletion layer _____.

(a) remains the same (b) is decreased (c) is increased (d) none of the above

21. A zener diode has _____.

(a) one pn junction (b) two pn junctions (c) three pn junctions (d) none of the above

22. A zener diode is used as _____.

(a) an amplifier (b) a voltage regulator (c) a rectifier (d) a multivibrator

23. The doping level in a zener diode is _____ that of a crystal diode.

(a) the same as (b) less than (c) more than (d) none of the above

24. A zener diode is always _____ connected.

(a) reverse (b) forward

(c) either reverse or forward (d) none of the above

25. A zener diode utilises _____ characteristic for its operation.

(a) forward (b) reverse

(c) both forward and reverse (d) none of the above

26. In the breakdown region, a zener diode behaves like a _____ source.

(a) constant voltage (b) constant current (c) constant resistance (d) none of the above

27. A zener diode is destroyed if it _____.

(a) is forward biased (b) is reverse biased

(c) carries more than rated current (d) none of the above

28. A series resistance is connected in the zener circuit to _____.

(a) properly reverse bias the zener (b) protect the zener

(c) properly forward bias the zener (d) none of the above

29. A zener diode is _____ device.

(a) a non-linear (b) a linear (c) an amplifying (d) none of the above

30. A zener diode has _____ breakdown voltage.

(a) undefined (b) sharp (c) zero (d) none of the above

31. _____ rectifier has the lowest forward resistance.

(a) solid state (b) vacuum tube (c) gas tube (d) none of the above

32. Mains a.c. power is converted into d.c. power for _____.

(a) lighting purposes (b) heaters

(c) using in electronic equipment (d) none of the above

33. The disadvantage of a half-wave rectifier is that the _____.

(a) components are expensive (b) diodes must have a higher power rating

(c) output is difficult to filter (d) none of the above

34. If the a.c. input to a half-wave rectifier has an r.m.s. value of $400/\sqrt{2}$ volts, then diode PIV rating is _____.

(a) $400/\sqrt{2}$ V (b) 400 V (c) $400 \times \sqrt{2}$ V (d) none of the above

35. The ripple factor of a half-wave rectifier is _____.

(a) 2 (b) 1.21 (c) 2.5 (d) 0.48

36. There is a need of transformer for _____.

(a) half-wave rectifier (b) centre-tap full-wave rectifier

(c) bridge full-wave rectifier (d) none of the above

37. The PIV rating of each diode in a bridge rectifier is _____ that of the equivalent centretap rectifier.

(a) one-half (b) the same as (c) twice (d) four times

38. For the same secondary voltage, the output voltage from a centre-tap rectifier is _____ than that of bridge rectifier.

(a) twice (b) thrice (c) four times (d) one-half

39. If the PIV rating of a diode is exceeded, _____.

(a) the diode conducts poorly (b) the diode is destroyed

(c) the diode behaves as zener diode (d) none of the above

40. A 10 V power supply would use _____ as filter capacitor.

(a) paper capacitor (b) mica capacitor (c) electrolytic capacitor (d) air capacitor

41. A 1000 V power supply would use _____ as a filter capacitor.

(a) paper capacitor (b) air capacitor (c) mica capacitor (d) electrolytic capacitor

42. The _____ filter circuit results in the best voltage regulation.

(a) choke input (b) capacitor input (c) resistance input (d) none of the above

43. A half-wave rectifier has an input voltage of 240 V_{rms}. If the step-down transformer has a turns ratio of 8 : 1, what is the peak load voltage ? Ignore diode drop.

(a) 27.5 V (b) 86.5 V (c) 30 V (d) 42.5 V

44. The maximum efficiency of a half-wave rectifier is _____.

(a) 40.6% (b) 81.2% (c) 50% (d) 25%

45. The most widely used rectifier is _____.

(a) half-wave rectifier (b) centre-tap full-wave rectifier

(c) bridge full-wave rectifier (d) none of the above

46. When a transistor is driven to saturation, ideally the output is _____.

(a) V_{CC} (b) 0 (c) $V_{CC}/2$ (d) $2\,V_{CC}$

47. The maximum speed of electronic switch can be _____ operations per second.

(a) 10^4 (b) 10 (c) 1000 (d) 10^9

48. When a transistor is driven to cut off, ideally the output is _____.

(a) V_{CC} (b) 0 (c) $V_{CC}/2$ (d) $V_{CC}/3$

49. _____ multivibrator is a square wave oscillator.

(a) monostable (b) astable (c) bistable (d) none of the above

50. An astable multivibrator has _____.

(a) one stable state (b) two stable states (c) no stable state (d) none of the above

51. If d.c. supply of 10 V is fed to a differentiating circuit, then output will be _____.

(a) 20 V (b) 10 V (c) 0 V (d) none of the above

52. If the input to a differentiating circuit is a saw-tooth wave, then output will be _____ wave.

(a) square (b) triangular (c) sine (d) rectangular

53. A bistable multivibrator has _____.

(a) two stable states (b) one stable state (c) no stable state (d) none of the above

54. If a square wave is fed to a differentiating circuit, the output will be _____.

(a) sine wave (b) sharp narrow pulses (c) rectangular wave (d) triangular wave

55. An integrating circuit is a simple RC series circuit with output taken across _____.

(a) both R and C (b) R (c) C (d) none of the above

56. For an integrating circuit to be effective, the RC product should be _____ the time period of the input wave.

(a) 5 times greater than (b) 5 times smaller than

(c) equal to (d) atleast 10 times greater than

57. A differentiating circuit is a simple RC circuit with output taken across _____.

(a) R (b) C (c) both R and C (d) none of the above

58. A monostable multivibrator has _____.

(a) no stable state (b) one stable state (c) two stable states (d) none of the above

59. The multivibrator which generates square wave of its own is the _____ multivibrator.

(a) monostable (b) bistable (c) astable (d) none of the above

60. For a differentiating circuit to be effective, the RC product should be _____ the time period of the input wave.

(a) equal to (b) 5 times greater than

(c) 5 times smaller than (d) atleast 10 times greater than

61. When a rectangular voltage waveform is applied to a capacitor, then the current waveform is _____.

(a) rectangular (b) sinusoidal (c) sawtooth (d) square

62. The positive clipper is that which removes the _____ half-cycles of the input voltage.

(a) negative (b) positive

(c) both positive and negative (d) none of the above

63. A clamping circuit adds _____ component to the signal.

(a) d.c. (b) a.c. (c) both d.c. and a.c. (d) none of the above

64. One would find a clamping circuit in _____.

(a) receiving antenna (b) radio transmitter (c) radio receiver (d) television receiver

65. When transistor is used as an amplifier, it is operated in the _____ region.

(a) off (b) saturation (c) active (d) none of the above

66. When the transistor (CE arrangement) is in the cut off region, the collector current is _____.

(a) I_{CBO} (b) I_{CEO} (c) $(\beta + 1)\, I_{CEO}$ (d) $I_{C\,(sat)}$

67. A negative clipper removes the _____ halfcycles of the input voltage.

(a) negative (b) positive

(c) both positive and negative (d) none of the above

68. If the input to an integrating circuit is a succession of alternating positive and negative pulses of very short duration, the output will be _____ wave.

(a) rectangular (b) triangular (c) sine (d) square

69. In a multivibrator, we have _____ feedback.

(a) negative
(b) 100 % positive
(c) both positive and negative
(d) none of the above

ANSWERS

1. (a) one pn junction
2. (b) Ω
3. (a) forward
4. (c) μA
5. (d) 0.7 V
6. (b) a rectifier
7. (c) less than
8. (a) conductor
9. (d) 40000 : 1
10. (a) minority carriers
11. (c) increases
12. (b) lower than
13. (c) is decreased
14. (d) barrier potential
15. (a) linear
16. (a) forward
17. (a) non-linear
18. (b) forward
19. (d) PIV rating
20. (c) is increased
21. (a) one pn junction
22. (b) a voltage regulator
23. (c) more than
24. (a) reverse
25. (b) reverse
26. (a) constant voltage
27. (c) carries more than rated current
28. (b) protect the zener
29. (a) a non-linear
30. (b) sharp
31. (a) solid state
32. (c) using in electronic equipment
33. (c) output is difficult to filter
34. (b) 400 V
35. (d) 0.48
36. (b) centre-tap full-wave rectifier
37. (a) one-half
38. (d) one-half
39. (b) the diode is destroyed
40. (c) electrolytic capacitor
41. (a) paper capacitor
42. (a) choke input
43. (d) 42.5 V
44. (a) 40.6%
45. (c) bridge full-wave rectifier
46. (b) 0
47. (d) 10^9
48. (a) V_{CC}
49. (b) astable
50. (c) no stable state
51. (c) 0 V
52. (d) rectangular
53. (a) two stable states
54. (b) sharp narrow pulses
55. (c) C
56. (d) atleast 10 times greater than
57. (a) R
58. (b) one stable state
59. (c) astable
60. (d) atleast 10 times greater than
61. (a) rectangular
62. (b) positive
63. (a) d.c.
64. (d) television receiver
65. (c) active
66. (b) I_{CEO}
67. (a) negative
68. (d) square
69. (b) 100 % positive

EXERCISE

1. Explain the intrinsic and extrinsic semiconductors.
2. Explain P type and N type semiconductors.
3. Explain the forward and reverse characteristic of PN junction diode.
4. Explain forward and reverse biasing of PN junction diode.
5. Draw the circuit of a half-wave rectifier and explain the working.
6. Draw the circuit and explain the working of full wave rectifier with two diodes. Draw the waveforms at the input and output.
7. Draw the circuit of full wave bridge rectifier and explain the working.
8. Explain π filter.
9. Compare the full wave rectifier with two diodes and a full wave bridge rectifier.
10. Derive the equation for I_{dc} and I_{RMS} for a full wave rectifier and from this obtain the ripple factor.
11. Obtain the expression for rectification efficiency.
12. What is PIV ? Explain its importance in rectifier circuits. State its value for a full wave bridge rectifier.
13. Compare the three types of rectifier circuits.
14. Explain the necessity of filters in rectifier circuits. State different types of filter circuits.
15. Compare choke input filter and capacitor input filter.
16. Explain important ratings of diode.
17. Define ripple voltage and percentage ripple.
18. Draw and explain positive and negative clipping circuit using diode.
19. Explain the difference between clipping and clamping circuits.
20. Draw the circuit and explain the working of voltage tripler.
21. Draw the characteristic of zener diode and explain how this characteristic is useful in voltage regulation.
22. Discuss the reasons of changes in the output voltage of a rectifier.
23. Draw the block diagram of an electronic voltage regulator and explain the function of each block.
24. Define and explain load regulation and line regulation with reference to rectifier.
25. Explain zener diode voltage regulator.
26. Draw the circuit of a shunt regulator and explain its working.
27. Draw the circuit of a simple series type voltage regulator circuit and explain its working.
28. State limitations of zener voltage regulator.
29. Draw the circuit of dual power supply using zener diodes.
30. Explain principle of SMPS.
31. Draw block diagram of SMPS and explain its working.
32. Explain uninterruptable power supplies.

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